जब तक आपको यह परीक्षण पुस्तिका खोलने को न कहा जाए तब तक न खोलें

T.B.C.: FIAS-SFG26 **परीक्षण पुस्तिका अनुक्रम**

परीक्षण पुस्तिका

सामान्य अध्ययन

A

पूर्णांक : 100 **समय: एक घंटे**

अनुदेश

1. परीक्षा प्रारम्भ होने के तुरन्त बाद, आप इस परीक्षण पुस्तिका की पड़ताल अवश्य कर लें कि इसमें कोई बिना छपा, फटा या छूटा हुआ पृष्ठ अथवा प्रश्नांश, आदि न हो। यदि ऐसा है, तो इसे सही परीक्षण पुस्तिका से बदल लीजिए।
2. **कृपया ध्यान रखें कि OMR उत्तर-पत्रक में, उचित स्थान पर, रोल नम्बर और परिक्षण पुस्तिका अनुक्रम A, B, C या D को, ध्यान से एवं बिना किसी चूक या विसंगति के भरने और कूटबद्ध करने की जिम्मेदारी उम्मीदवार की है। किसी भी प्रकार की चूक/विसंगति की स्थिति में उत्तर -पत्रक निरस्त कर दिया जाएगा ।**
3. इस परीक्षण पुस्तिका पर साथ में दिए गए कोष्ठक में आपको अपना अनुक्रमांक लिखना है। परीक्षण पुस्तिका पर **और कुछ न** लिखें।

नाम:
ईमेल:
मोबाइल:

4. इस परीक्षण पुस्तिका में **50** प्रश्नांश (प्रश्न) हैं। प्रत्येक प्रश्नांश **अंग्रेज़ी और हिंदी** में छपा है। प्रत्येक प्रश्नांश में चार प्रत्युत्तर (उत्तर) दिए गए हैं। इनमें से एक प्रत्युतर को चुन लें, जिसे आप उत्तर-पत्रक पर अंकित करना चाहते हैं। यदि आपको ऐसा लगे कि एक से अधिक प्रत्युत्तर सही हैं, तो उस प्रत्युत्तर को अंकित करें जो आपको सर्वोत्तम लगे। किसी भी स्थिति में, प्रत्येक प्रश्नांश के लिए **केवल एक ही** प्रत्युत्तर चुनना है।
5. आपको अपने सभी प्रत्युत्तर अलग से दिए गए उत्तर-पत्रक पर ही अंकित करने हैं। उत्तर-पत्रक में दिए गए निर्देश देखिए।
6. ***सभी*** प्रश्नांशों *के अंक समान हैं।*
7. इससे पहले कि आप परीक्षण पुस्तिका के विभिन्न प्रश्नांशों के प्रत्युत्तर उत्तर-पत्रक पर अंकित करना शुरू करें, आपको प्रवेश प्रमाण-पत्र के साथ प्रेषित अनुदेशों के अनुसार कुछ विवरण उत्तर-पत्रक में देने हैं।
8. आप अपने सभी प्रत्युत्तरों को उत्तर-पत्रक में भरने के बाद तथा परीक्षा के समापन पर केवल उत्तर-पत्रक अधीक्षक को सौंप दें। आपको अपने साथ परीक्षण पुस्तिका ले जाने की अनुमति है।
9. कच्चे काम के लिए पत्रक परीक्षण पुस्तिका के अंत में संलग्न है।
10. **गलत उत्तरों के लिए दंड :**

वस्तुनिष्ठ प्रश्न-पत्रों में उम्मीदवार द्वारा दिए गए गलत उत्तरों के लिए दंड दिया जाएगा।

(i) प्रत्येक प्रश्न के लिए चार वैकल्पिक उत्तर हैं। उम्मीदवार द्वारा प्रत्येक प्रश्न के लिए दिए गए एक गलत उत्तर के लिए प्रश्न हेतु नियत किए गए अंकों का एक-तिहाई दंड के रूप में काटा जाएगा।

(ii) यदि कोई उम्मीदवार एक से अधिक उत्तर देता है, तो इसे गलत उत्तर माना जाएगा, यद्यपि दिए गए उत्तरों में से एक उत्तर सही होता है, फिर भी उस प्रश्न के लिए उपर्युक्तानुसार ही, उसी तरह का दंड दिया जाएगा।

(iii) यदि उम्मीदवार द्वारा कोई प्रश्न हल नहीं किया जाता है, अर्थात उम्मीदवार द्वारा उत्तर नहीं दिया जाता है, तो उस प्रश्न के लिए कोई दंड नहीं दिया जाएगा।

नोट: प्रश्न का उत्तर दें तथा अभ्यर्थी आत्मविश्वास स्तर (100% / 75% / 50% / 0%) का उपयोग करके उत्तर के प्रति अपनी निश्चितता का स्तर दर्ज कर सकते हैं। आप प्रत्येक प्रश्न के नीचे उपयुक्त आत्मविश्वास स्तर का चयन कर सकते हैं। इससे उन प्रश्नों की पुनरावृत्ति (रिवीजन) में सहायता मिलेगी, जिनमें उत्तर सही होने के बावजूद आप अनिश्चित थे।

जब तक आपको यह परीक्षण पुस्तिका खोलने को न कहा जाए तब तक न खोलें

Q.1) Consider the following statements:

Statement I: Article 6 of the Paris Agreement on climate change is frequently discussed in global discussions on sustainable development and climate change.

Statement II: Article 6 of the Paris Agreement on climate change sets out the principles of carbon markets.

Statement III: Article 6 of the Paris Agreement on climate change intends to promote inter-country non-market strategies to reach their climate targets.

Which one of the following is correct in respect of the above statements?

a) Both Statement II and Statement III are correct and both of them explain Statement I
b) Both Statement II and Statement III are correct but only one of them explains Statement I
c) Only one of the Statements II and III is correct and that explains Statement I
d) Neither Statement II nor Statement III is correct

100%		75%		50%		0%	

Q.2) Consider the following International Environmental convention/protocol:

I. Rotterdam Convention
II. Montreal Protocol
III. Stockholm Convention
IV. Basel Convention

With reference to the convention/protocol mentioned above, which one of the following is the correct chronology of the years in which they entered into force?

a) II - IV - III - I
b) II - IV - I -III
c) IV - II - I - III
d) IV - II - III - I

100%		75%		50%		0%	

Q.3) With reference to the Carbon Credit Trading Scheme (CCTS) in India, consider the following statements:

I. Each carbon credit certificate issued will represent a hundred tons of CO_2 equivalent (tCO_2e) reduction (or removal) from the atmosphere.
II. It operates through two mechanisms that is the Compliance Mechanism and the Offset Mechanism.
III. The Central Electricity Regulatory Commission is the agency responsible for issuing carbon credit certificates.

Which of the statements given above is/are correct?

a) I and II only
b) II only
c) II and III only
d) None of the above

100%		75%		50%		0%	

Q.4) The "Keeling Curve", a significant indicator in climate science, is associated with the long-term monitoring of which of the following?

a) Glacial retreat in Arctic and Antarctic regions
b) Ocean acidity
c) Atmospheric carbon dioxide (CO_2) concentrations
d) Stratospheric ozone depletion

100%		75%		50%		0%	

Q.5) Consider the following statements with respect to the Climate and Clean Air Coalition (CCAC):

I. Its primary objective is to reduce long-lived climate pollutants i.e. those pollutants that remain in the atmosphere for a much longer period of time than carbon dioxide.
II. It mandates binding emission reduction targets for its member countries.

Which of the statements given above is/are correct?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.1) निम्नलिखित कथनों पर विचार कीजिए:

कथन I: जलवायु परिवर्तन पर पेरिस समझौते का अनुच्छेद 6 सतत विकास और जलवायु परिवर्तन पर वैश्विक चर्चाओं में अक्सर उठाया जाता है।

कथन II: जलवायु परिवर्तन पर पेरिस समझौते का अनुच्छेद 6 कार्बन बाजारों के सिद्धांतों को निर्धारित करता है।

कथन III: जलवायु परिवर्तन पर पेरिस समझौते का अनुच्छेद 6 अपने जलवायु लक्ष्यों को प्राप्त करने के लिए अंतर-देशीय गैर-बाजार रणनीतियों को बढ़ावा देने का इरादा रखता है।

उपर्युक्त कथनों के संबंध में निम्नलिखित में से कौन सा सही है?

a) कथन II और कथन III दोनों सही हैं और दोनों ही कथन I की व्याख्या करते हैं।
b) कथन II और कथन III दोनों सही हैं लेकिन उनमें से केवल एक ही कथन I की व्याख्या करता है।
c) कथन II और III में से केवल एक ही सही है और वह कथन I की व्याख्या करता है।
d) न तो कथन II और न ही कथन III सही है।

100%		75%		50%		0%	

Q.2) निम्नलिखित अंतर्राष्ट्रीय पर्यावरण सम्मेलन/प्रोटोकॉल पर विचार कीजिए:

I. रॉटरडैम सम्मेलन
II. मॉन्ट्रियल प्रोटोकॉल
III. स्टॉकहोम सम्मेलन
IV. बेसल सम्मेलन

उपर्युक्त सम्मेलन/प्रोटोकॉल के संदर्भ में, निम्नलिखित में से कौन सा उनके लागू होने के वर्षों का सही क्रम है?

a) II - IV - III - I
b) II - IV - I - III
c) IV - II - I - III
d) IV - II - III - I

100%		75%		50%		0%	

Q.3) भारत में कार्बन क्रेडिट ट्रेडिंग स्कीम (CCTS) के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. जारी किया गया प्रत्येक कार्बन क्रेडिट प्रमाणपत्र वायुमंडल से सौ टन CO_2 समतुल्य (tCO_2e) की कमी (या निष्कासन) का प्रतिनिधित्व करता है।
II. यह दो तंत्रों के माध्यम से संचालित होता है, अर्थात अनुपालन तंत्र और ऑफसेट तंत्र।
III. केंद्रीय विद्युत नियामक आयोग कार्बन क्रेडिट प्रमाणपत्र जारी करने के लिए जिम्मेदार एजेंसी है।

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?

a) केवल I और II
b) केवल II
c) केवल II और III
d) उपर्युक्त में से कोई नहीं

100%		75%		50%		0%	

Q.4) जलवायु विज्ञान में एक महत्वपूर्ण सूचक, "कीलिंग वक्र", निम्नलिखित में से किसके दीर्घकालिक निगरानी से संबंधित है?

a) आर्कटिक और अंटार्कटिक क्षेत्रों में हिमनदों का पीछे हटना
b) महासागरों की अम्लता
c) वायुमंडलीय कार्बन डाइऑक्साइड (CO_2) की सांद्रता
d) समताप मंडल में ओजोन परत का क्षरण

100%		75%		50%		0%	

Q.5) जलवायु और स्वच्छ वायु गठबंधन (CCAC) के संबंध में निम्नलिखित कथनों पर विचार कीजिए:

I. इसका प्राथमिक उद्देश्य दीर्घकालिक जलवायु प्रदूषकों को कम करना है अर्थात वे प्रदूषक जो कार्बन डाइऑक्साइड की तुलना में वायुमंडल में बहुत लंबे समय तक बने रहते हैं।
II. यह अपने सदस्य देशों के लिए बाध्यकारी उत्सर्जन कटौती लक्ष्य निर्धारित करता है।

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?

a) केवल I
b) केवल II
c) I और II दोनों
d) न तो I और न ही II

100%		75%		50%		0%	

Q.6) The Montreal Protocol on Substances that Deplete the Ozone Layer is a global agreement to protect the Earth's ozone layer by phasing out the chemicals that deplete it. In this context, this phase-out plan includes:

I. Production of ozone-depleting substances.
II. Consumption of ozone-depleting substances.

Which of the statements given above is/are correct?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.7) With reference to the 'Wet-bulb temperature', consider the following statements:

I. It is the lowest temperature of air that can be reached by evaporating water into the air, at constant pressure.
II. High wet-bulb temperatures can limit the human body's ability to cool itself through sweating.
III. Higher humidity leads to a greater difference between dry-bulb and wet-bulb temperatures.

Which of the statements given above are correct?

a) I and II only
b) II and III only
c) I and III only
d) I, II and III

100%		75%		50%		0%	

Q.8) In the context of climate studies, the term 'Snow Persistence' measures the:

a) fraction of time snow remains on the ground after snowfall.
b) total depth of snow accumulated on a mountain peak during a single winter season.
c) reflectivity of snow-covered surfaces influencing Earth's energy balance.
d) process of transformation of snow into glacial ice over time.

100%		75%		50%		0%	

Q.9) With reference to Ocean Darkening, consider the following statements:

I. It is the shrinking of the photic zones or those layers of water where sunlight can pass and induce the process of photosynthesis
II. Increased sediment runoff from rivers and coastal erosion contributes to ocean darkening.

Which of the statements given above is/are correct?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.10) In the context of the financial sector, which of the following best describes the term 'Co-lending':

a) A system in which two borrowers jointly apply for a single loan to reduce their individual credit burden.
b) A credit arrangement in which two banks compete to offer a lower interest rate to the same borrower.
c) A financing arrangement where a bank and a Non-Banking Financial Company (NBFC) jointly provide a loan and share funds and risks.
d) A process where a bank refinances loans already issued by an NBFC to improve the NBFC's liquidity.

100%		75%		50%		0%	

Q.11) Which one of the following statements best describes the term 'Social Cost of Carbon'? It is a measure, in monetary value, of the

a) long-term damage done by a tonne of CO_2 emissions in a given year.
b) requirement of fossil fuels for a country to provide goods and services to its citizens, based on the burning of those fuels.
c) efforts put in by a climate refugee to adapt to live in a new place.
d) contribution of an individual person to the carbon footprint on the planet Earth.

100%		75%		50%		0%	

Q.6) ओजोन परत को नष्ट करने वाले पदार्थों पर मॉन्ट्रियल प्रोटोकॉल, पृथ्वी की ओजोन परत को नष्ट करने वाले रसायनों को चरणबद्ध तरीके से समाप्त करके उसकी रक्षा करने का एक वैश्विक समझौता है। इस संदर्भ में, इस चरणबद्ध समाप्ति योजना में शामिल हैं:

I. ओजोन-क्षयकारी पदार्थों का उत्पादन।

II. ओजोन-क्षयकारी पदार्थों का उपभोग।

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?

a) केवल I

b) केवल II

c) I और II दोनों

d) न तो I और न ही II

100%		75%		50%		0%	

Q.7) वेट बल्ब तापमान' के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. यह वायु का वह न्यूनतम तापमान है जो स्थिर दाब पर पानी के वाष्पीकरण द्वारा प्राप्त किया जा सकता है।

II. उच्च वेट बल्ब तापमान पसीने के माध्यम से शरीर को ठंडा करने की क्षमता को सीमित कर सकता है।

III. उच्च आर्द्रता ड्राई बल्ब और वेट बल्ब तापमान के बीच अधिक अंतर का कारण बनती है।

उपर्युक्त कथनों में से कौन से सही हैं?

a) केवल I और II

b) केवल II और III

c) केवल I और III

d) I, II और III

100%		75%		50%		0%	

Q.8) जलवायु अध्ययन के संदर्भ में, 'हिम स्थायित्व (Snow Persistence)' शब्द निम्नलिखित को मापता है:

a) हिमपात के बाद बर्फ़ के ज़मीन पर बने रहने का समय।

b) एक शीतकालीन ऋतु के दौरान किसी पर्वत शिखर पर संचित बर्फ की कुल गहराई।

c) बर्फ़ से ढकी सतहों की परावर्तन शक्ति जो पृथ्वी के ऊर्जा संतुलन को प्रभावित करती है।

d) समय के साथ बर्फ का हिमनदीय बर्फ (ग्लेशियल आइस) में परिवर्तित होने की प्रक्रिया।

100%		75%		50%		0%	

Q.9) महासागरीय अंधकार (ओशन डार्कनिंग) के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. यह प्रकाशीय क्षेत्रों का संकुचन है, यानी जल की वे परतें जहाँ से सूर्य का प्रकाश गुजर सकता है और प्रकाश संश्लेषण की प्रक्रिया को प्रेरित कर सकता है।

II. नदियों से बढ़ते तलछट अपवाह और तटीय अपरदन महासागरीय अंधकार (ओशन डार्कनिंग) में योगदान करते हैं।

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?

a) केवल I

b) केवल II

c) I और II दोनों

d) न तो I और न ही II

100%		75%		50%		0%	

Q.10) वित्तीय क्षेत्र के संदर्भ में, निम्नलिखित में से कौन सा विकल्प 'सह-ऋण (को-लेंडिंग)' शब्द का सबसे सटीक वर्णन करता है:

a) एक ऐसी प्रणाली जिसमें दो उधारकर्ता अपने व्यक्तिगत ऋण भार को कम करने के लिए संयुक्त रूप से एक ही ऋण के लिए आवेदन करते हैं।

b) एक ऋण व्यवस्था जिसमें दो बैंक एक ही उधारकर्ता को कम ब्याज दर देने के लिए प्रतिस्पर्धा करते हैं।

c) एक वित्तपोषण व्यवस्था जिसमें एक बैंक और एक गैर-बैंकिंग वित्तीय कंपनी (NBFC) संयुक्त रूप से ऋण प्रदान करते हैं और धन और जोखिम साझा करते हैं।

d) एक प्रक्रिया जिसमें एक बैंक, NBFC की तरलता में सुधार करने के लिए, NBCF द्वारा पहले से जारी किए गए ऋणों का पुनर्वित्त करता है।

100%		75%		50%		0%	

Q.11) निम्नलिखित में से कौन सा कथन 'कार्बन की सामाजिक लागत' को सबसे अच्छी तरह से परिभाषित करता है?

यह मौद्रिक मूल्य में एक माप है, जो निम्न को दर्शाता है:

a) किसी दिए गए वर्ष में एक टन CO_2 उत्सर्जन से होने वाला दीर्घकालिक नुकसान।

b) किसी देश द्वारा अपने नागरिकों को वस्तुओं और सेवाओं की आपूर्ति के लिए जीवाश्म ईंधन की आवश्यकता, जो इन ईंधनों के जलने पर आधारित है।

c) किसी जलवायु शरणार्थी द्वारा नए स्थान पर रहने के लिए किए गए प्रयास।

d) पृथ्वी ग्रह पर कार्बन पदचिह्न में किसी व्यक्ति का योगदान।

100%		75%		50%		0%	

Q.12) In the context of the Paris Agreement, consider the following statements about the Biennial Transparency Report (BTR):

I. Only the developed countries that are parties to the Paris agreement are required to submit the report every two years.

II. It is part of the Enhanced Transparency framework of the Paris Agreement and replaces the earlier 'Biennial Update Reports' and 'Biennial Reports' submitted by countries.

Which of the statements given above is/are correct?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.13) The triple planetary crisis refers to the three main interlinked issues that humanity currently faces. It includes:

a) climate change, biodiversity loss, and pollution.
b) climate change, economic recession, and political instability.
c) biodiversity loss, desertification, and food insecurity.
d) pollution, global conflict, and migration.

100%		75%		50%		0%	

Q.14) It is one of the widest glaciers in the world, spanning approximately 120 km. Often referred to as the 'Doomsday Glacier,' its rapid melting is a major concern for global sea-level rise, as its collapse could eventually trigger the loss of a significant portion of the ice sheet.

Which of the following glaciers is being described above?

a) Lambert Glacier
b) Yala Glacier
c) Thwaites Glacier
d) Jakobshavn Glacier

100%		75%		50%		0%	

Q.15) Consider the following:

I. A significant rise in the temperature in the regions of North Western Europe

II. Alter the timing of Indian Monsoons

III. A rise in sea levels along the North American Atlantic coast

IV. A global increase in the rate of ocean carbon sequestration

How many of the above are possible consequences of the weakening of Atlantic Meridional Overturning Circulation (AMOC)?

a) Only one
b) Only two
c) Only three
d) All the four

100%		75%		50%		0%	

Q.16) With reference to hydrogen production, hydrogen produced through electrolysis of water using electricity generated from nuclear energy is known as:

a) Pink Hydrogen
b) Turquoise Hydrogen
c) White Hydrogen
d) Black Hydrogen

100%		75%		50%		0%	

Q.17) With reference to the ecological impacts of marine heat waves, consider the following statements:

I. It may promote the growth of harmful algal blooms in coastal waters.

II. It may cause large-scale dieback of seagrass meadows.

III. It can alter the migration pattern of marine species.

IV. It will boost the growth of Kelp forests.

Which of the statements given above is/are correct?

a) I and II only
b) I, II and III only
c) III and IV only
d) I, II, III and IV

100%		75%		50%		0%	

Q.12) पेरिस समझौते के संदर्भ में, द्विवार्षिक पारदर्शिता रिपोर्ट (BTR) के बारे में निम्नलिखित कथनों पर विचार कीजिए:

I. केवल पेरिस समझौते के पक्षकार विकसित देशों को ही हर दो साल में रिपोर्ट प्रस्तुत करना अनिवार्य है।

II. यह पेरिस समझौते के उन्नत पारदर्शिता ढांचे का हिस्सा है और देशों द्वारा प्रस्तुत की जाने वाली पूर्व की 'द्विवार्षिक अद्यतन रिपोर्ट' और 'द्विवार्षिक रिपोर्ट' का स्थान लेती है।

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?

a) केवल I

b) केवल II

c) I और II दोनों

d) न तो I और न ही II

100%		75%		50%		0%	

Q.13) त्रिपक्षीय ग्रहीय संकट से तात्पर्य मानवता के सामने मौजूद तीन मुख्य परस्पर जुड़े मुद्दों से है। इनमें शामिल हैं:

a) जलवायु परिवर्तन, जैव विविधता का क्षरण और प्रदूषण।

b) जलवायु परिवर्तन, आर्थिक मंदी और राजनीतिक अस्थिरता।

c) जैव विविधता का क्षरण, मरुस्थलीकरण और खाद्य असुरक्षा।

d) प्रदूषण, वैश्विक संघर्ष और प्रवासन।

100%		75%		50%		0%	

Q.14) यह दुनिया के सबसे चौड़े ग्लेशियरों में से एक है, जो लगभग 120 किमी तक फैला हुआ है। इसे अक्सर 'डूम्सडे ग्लेशियर' कहा जाता है, और इसका तेजी से पिघलना वैश्विक समुद्र स्तर में वृद्धि के लिए एक बड़ी चिंता का विषय है, क्योंकि इसके विघटित होने की स्थिति में अंततः हिमावरण का एक महत्वपूर्ण हिस्सा नष्ट हो सकता है।

उपर्युक्त वर्णित ग्लेशियर निम्नलिखित में से कौन सा है?

a) लैम्बर्ट ग्लेशियर

b) याला ग्लेशियर

c) थ्वाइट्स ग्लेशियर

d) जैकबसन ग्लेशियर

100%		75%		50%		0%	

Q.15) निम्नलिखित पर विचार कीजिए:

I. उत्तर पश्चिमी यूरोप के क्षेत्रों में तापमान में महत्वपूर्ण वृद्धि

II. भारतीय मानसून के समय में परिवर्तन

III. उत्तरी अमेरिकी अटलांटिक तट पर समुद्र स्तर में वृद्धि

IV. महासागर कार्बन पृथक्करण की दर में वैश्विक वृद्धि

उपर्युक्त में से कितने अटलांटिक मेरिडियनल ओवरटर्निंग सर्कुलेशन (AMOC) के कमजोर होने के संभावित परिणाम हैं?

a) केवल एक

b) केवल दो

c) केवल तीन

d) सभी चार

100%		75%		50%		0%	

Q.16) हाइड्रोजन उत्पादन के संदर्भ में, परमाणु ऊर्जा से उत्पन्न बिजली का उपयोग करके जल के विद्युत अपघटन द्वारा उत्पादित हाइड्रोजन को निम्न नाम से जाना जाता है:

a) पिंक हाइड्रोजन

b) टर्कॉइज़ हाइड्रोजन

c) व्हाइट हाइड्रोजन

d) ब्लैक हाइड्रोजन

100%		75%		50%		0%	

Q.17) समुद्री हीटवेव के पारिस्थितिक प्रभावों के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. यह तटीय जल में हानिकारक शैवाल प्रस्फुटन को बढ़ावा दे सकता है।

II. यह समुद्री घास के मैदानों के बड़े पैमाने पर सूखने का कारण बन सकता है।

III. यह समुद्री प्रजातियों के प्रवास पैटर्न को बदल सकता है।

IV. यह केल्प वनों के विकास को बढ़ावा देगा।

उपर्युक्त कथनों में से कौन सा/से सही है/हैं?

a) केवल I और II

b) केवल I, II और III

c) केवल III और IV

d) I, II, III और IV

100%		75%		50%		0%	

Q.18) Which of the following is/are causes of coral bleaching?
I. Decreased solar radiation
II. Ocean alkalinization
III. Extreme low tides
IV. Nutrient enrichment of oceanic water
Select the correct answer using the code given below:
a) I and II only
b) IV only
c) III and IV only
d) II, III and IV

100%		75%		50%		0%	

Q.19) With reference to environmental governance, which one of the following is related to the concept of 'GreenRinsing'?
a) A practice where firms rely on collective industry commitments to substitute for individual sustainability action.
b) A practice where firms are adopting and later abandoning ambitious net-zero targets to secure investments.
c) A practice where environmental organisations collectively lobby corporations to adopt net-zero pledges.
d) A practice where companies overstate environmental benefits using sector-wide aggregate data.

100%		75%		50%		0%	

Q.20) Consider the following statements:
I. India is the world's top producer of aluminium globally.
II. Rajasthan holds the highest share of copper reserves in India.
III. Chile has the largest known copper reserves in the world.
Which of the statements given above are correct?
a) I and II only
b) I and III only
c) II and III only
d) I, II and III

100%		75%		50%		0%	

Q.21) Consider the following statements:
Statement I: Scientific studies suggest that a shift is taking place in the Earth's rotation and axis.
Statement II: Solar flares and associated coronal mass ejections bombarded the Earth's outermost atmosphere with a tremendous amount of energy.
Statement III: As the Earth's polar ice melts, the water tends to move towards the equator.
Which one of the following is correct in respect of the above statements?
a) Both Statement II and Statement III are correct and both of them explain Statement I
b) Both Statement II and Statement III are correct but only one of them explains Statement I
c) Only one of the Statements II and III is correct and that explains Statement I
d) Neither Statement II nor Statement III is correct

100%		75%		50%		0%	

Q.22) With reference to the Aerosols, consider the following statements:
I. Sulfate and nitrate aerosols have a net cooling effect on the climate by reflecting incoming solar radiation back into space.
II. Black carbon aerosols near the surface absorb heat and warm the surrounding air
III. Rapidly reducing aerosol pollution without simultaneous reduction in greenhouse gases can lead to an increase in atmospheric temperature
Which of the statements given above are correct?
a) I and II only
b) II and III only
c) I and III only
d) I, II and III

100%		75%		50%		0%	

Q.23) Which of the following factors primarily contribute to the occurrence of 'thirst waves'?
I. Rapidly rising temperatures.
II. Consistently low atmospheric humidity.
III. Increased wind speed.
IV. High soil moisture levels.
Select the correct answer using the code given below:
a) I, II and IV only
b) III and IV only
c) I, II and III only
d) I, II, III and IV

100%		75%		50%		0%	

Q.18) निम्नलिखित में से कौन-सा/से प्रवाल विरंजन के कारण/हैं?
I. सौर विकिरण में कमी
II. महासागरीय क्षारीकरण
III. अत्यधिक निम्न ज्वार
IV. महासागरीय जल का पोषक तत्व संवर्धन
नीचे दिए गए कूट का उपयोग करके सही उत्तर का चयन कीजिए:
a) केवल I और II
b) केवल IV
c) केवल III और IV
d) II, III और IV

100%		75%		50%		0%	

Q.19) पर्यावरणीय शासन के संदर्भ में, निम्नलिखित में से कौन सा 'ग्रीनरिंसिंग' की अवधारणा से संबंधित है?
a) एक ऐसी प्रथा जिसमें कंपनियां व्यक्तिगत स्थिरता संबंधी कार्यों के स्थान पर सामूहिक उद्योग प्रतिबद्धताओं पर निर्भर करती हैं।
b) एक ऐसी प्रथा जिसमें कंपनियां निवेश हासिल करने के लिए महत्वाकांक्षी नेट-ज़ीरो लक्ष्य अपनाती हैं और बाद में उन्हें छोड़ देती हैं।
c) एक ऐसी प्रथा जिसमें पर्यावरण संगठन सामूहिक रूप से कंपनियों पर नेट-ज़ीरो प्रतिज्ञा अपनाने के लिए दबाव डालते हैं।
d) एक ऐसी प्रथा जिसमें कंपनियां क्षेत्र-व्यापी समग्र डेटा का उपयोग करके पर्यावरणीय लाभों को बढ़ा-चढ़ाकर पेश करती हैं।

100%		75%		50%		0%	

Q.20) निम्नलिखित कथनों पर विचार कीजिए:
I. भारत वैश्विक स्तर पर एल्यूमीनियम का दुनिया का शीर्ष उत्पादक है।
II. भारत में तांबे के भंडार का सबसे अधिक हिस्सा राजस्थान में है।
III. चिली में दुनिया में सबसे बड़ा ज्ञात तांबे का भंडार है।
उपर्युक्त कथनों में से कौन सा सही है?
a) केवल I और II
b) केवल I और III
c) केवल II और III
d) I, II और III

100%		75%		50%		0%	

Q.21) निम्नलिखित कथनों पर विचार कीजिए :
कथन I : वैज्ञानिक अध्ययन यह सुझाते हैं कि पृथ्वी के घूर्णन और अक्ष में परिवर्तन हो रहा है।
कथन II : सौर संस्फुरों (फ्लेयर्स) और उससे संबद्ध किरीटी द्रव्यमान निष्कासनों ने पृथ्वी के सबसे बाहरी वायुमंडल पर विशाल ऊर्जा-राशि की बौछार कर दी है।
कथन III : जैसे-जैसे पृथ्वी का ध्रुवीय हिम पिघलता है, जल की प्रवृत्ति विषुवत्-वृत्त की ओर जाने की होती है।
उपर्युक्त कथनों के बारे में, निम्नलिखित में से कौन-सा एक सही है?
a) कथन II और कथन III दोनों सही है तथा वे दोनों कथन I की व्याख्या करते हैं
b) कथन II और कथन III दोनों सही है किन्तु उनमें से केवल एक, कथन 1 की व्याख्या करता है
c) कथन II और कथन III में से केवल एक सही है तथा वह कथन I की व्याख्या करता है
d) न तो कथन II और न ही कथन III सही है

100%		75%		50%		0%	

Q.22) एरोसोल के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:
I. सल्फेट और नाइट्रेट एरोसोल आने वाले सौर विकिरण को अंतरिक्ष में वापस परावर्तित करके जलवायु पर शुद्ध शीतलन प्रभाव डालते हैं।
II. सतह के पास ब्लैक कार्बन एरोसोल गर्मी को अवशोषित करते हैं और आसपास की वायु को गर्म करते हैं
III. ग्रीनहाउस गैसों में एक साथ कमी के बिना एयरोसोल प्रदूषण को तेजी से कम करने से वायुमंडलीय तापमान में वृद्धि हो सकती है
उपर्युक्त कथनों में से कौन सा सही है?
a) केवल I और II
b) केवल II और III
c) केवल I और III
d) I, II और III

100%		75%		50%		0%	

Q.23) निम्नलिखित में से कौन सा कारक मुख्य रूप से 'थर्स्ट वेव्स'' की घटना में योगदान देता है?
I. तेजी से बढ़ता तापमान।
II. लगातार कम होती वायुमंडलीय आर्द्रता।
III. वायु की गति में वृद्धि।
IV. मृदा में नमी का उच्च स्तर।
नीचे दिए गए कूट का उपयोग करके सही उत्तर का चयन कीजिए:
a) केवल I, II और IV
b) केवल III और IV
c) केवल I, II और III
d) I, II, III और IV

100%		75%		50%		0%	

Q.24) Which of the following best describes the concept of Hydro-Climate Whiplash?
It refers to:
a) the rapid transition between two weather extremes, extremely wet weather and dangerously dry conditions.
b) the gradual long-term decline in average annual precipitation across semi-arid regions.
c) the phenomenon where increased glacial melt causes seasonal flooding and river channel displacement.
d) the intensification of monsoon variability across tropical regions resulting from sea surface temperature anomalies.

100%		75%		50%		0%	

Q.25) What is the Rio+20 Conference, often mentioned in the news?
a) It is the United Nations Conference on Sustainable Development
b) It is a Ministerial Meeting of the World Trade Organization
c) It is a Conference of the Inter Government Panel on Climate Change.
d) It is a Conference of the Member Countries of the Convention on Biological Diversity.

100%		75%		50%		0%	

Q.26) With reference to the 'Belém 4x Pledge', consider the following statements:
I. It aims to accelerate the clean energy transition through the adoption of clean energy sources, such as biogases and biofuels, as well as hydrogen and synthetic fuels.
II. The pledge is endorsed and supported by India.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.27) With reference to the Systematic Observation Impact Bond, recently seen in the news, consider the following statements:
I. It is the world's first impact bond aimed at strengthening weather and climate data systems globally.
II. It has been launched by the World Bank.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.28) With reference to the Paris Agreement under the United Nations Framework of Climate Change (UNFCCC), consider the following statements:
Statement I: The Paris Agreement seeks to find a Goldilocks solution between the contrasting models of the Kyoto Protocol and the 2009 Copenhagen Accord.
Statement II: The Paris Agreement is a treaty within the meaning of international law, but not all its provisions establish legal obligations.
Statement III: The Paris Agreement takes a bottom-up approach as it allows parties to nationally determine their contributions to address climate change.
Which one of the following is correct in respect of the above statements?
a) Both Statement II and Statement III are correct and both of them explain Statement I.
b) Both Statement II and Statement III are correct but only one of them explains Statement I.
c) Only one of the Statements II and III is correct and that explains Statement I.
d) Neither Statement II nor Statement III is correct.

100%		75%		50%		0%	

Q.24) निम्नलिखित में से कौन हाइड्रो-क्लाइमेट व्हिपलैश की अवधारणा का सबसे अच्छा वर्णन करता है?
यह संदर्भित करता है:

a) दो चरम मौसमी स्थितियों, अत्यधिक आर्द्र मौसम और अत्यंत गंभीर शुष्क परिस्थितियों के बीच तीव्र संक्रमण।
b) अर्ध-शुष्क क्षेत्रों में औसत वार्षिक वर्षा में क्रमिक दीर्घकालिक गिरावट को।
c) उस घटना को जहाँ ग्लेशियरों के पिघलने में वृद्धि के कारण मौसमी बाढ़ और नदी के प्रवाह मार्ग का विस्थापन होता है।
d) समुद्री सतह के तापमान की विसंगतियों के परिणामस्वरूप उष्णकटिबंधीय क्षेत्रों में मानसून की परिवर्तनशीलता के तीव्र होने को।

100%		75%		50%		0%	

Q.25) आम तौर पर समाचारों में आने वाला रियो+20 (Rio+20) सम्मेलन क्या है?

a) यह धारणीय विकास (सस्टेनेबल डेवलप्मेन्ट) पर संयुक्त राष्ट्र सम्मेलन है
b) यह विश्व व्यापार संगठन की मंत्रीवर्गीय (मिनिस्टीरियल) बैठक है
c) यह जलवायु परिवर्तन पर अन्तर-सरकारी पैनल (इंटर-गवर्नमेंटल पैनल ऑन क्लाइमेट चेंज) का सम्मेलन है
d) यह जैव विविधता पर कन्वेंशन के सदस्य देशों का सम्मेलन है

100%		75%		50%		0%	

Q.26) 'बेलेम 4x संकल्प' के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. इसका उद्देश्य स्वच्छ ऊर्जा स्रोतों, जैसे बायोगैस और जैव ईंधन, साथ ही हाइड्रोजन और सिंथेटिक ईंधन को अपनाने के माध्यम से स्वच्छ ऊर्जा संक्रमण में तेजी लाना है।
II. इस संकल्प' का भारत द्वारा समर्थन और सहयोग किया जाता है।

उपर्युक्त कथनों में से कौन-सा/से सही है/हैं?

a) केवल I
b) केवल II
c) I और II दोनों
d) न तो I और न ही II

100%		75%		50%		0%	

Q.27) हाल ही में समाचारों में देखे गए सिस्टमैटिक ऑब्जर्वेशन इम्पैक्ट बॉन्ड के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. यह दुनिया का पहला इम्पैक्ट बॉन्ड है जिसका उद्देश्य विश्व स्तर पर मौसम और जलवायु डेटा प्रणालियों को मजबूत करना है।
II. इसे विश्व बैंक द्वारा लॉन्च किया गया है।

उपर्युक्त कथनों में से कौन-सा/से सही है/हैं?

a) केवल I
b) केवल II
c) I और II दोनों
d) न तो I और न ही II

100%		75%		50%		0%	

Q.28) संयुक्त राष्ट्र जलवायु परिवर्तन फ्रेमवर्क (UNFCCC) के तहत पेरिस समझौते के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

कथन I: पेरिस समझौता, क्योटो प्रोटोकॉल और 2009 के कोपेनहेगन समझौते के परस्पर विरोधी मॉडलों के बीच एक 'गोल्डीलॉक्स समाधान' खोजने का प्रयास करता है।

कथन II: पेरिस समझौता अंतर्राष्ट्रीय कानून के अर्थ में एक संधि है, लेकिन इसके सभी प्रावधान कानूनी दायित्व निर्धारित नहीं करते हैं।

कथन III: पेरिस समझौता एक बॉटम-अप दृष्टिकोण अपनाता है क्योंकि यह पक्षों को जलवायु परिवर्तन को निपटने के लिए राष्ट्रीय स्तर पर अपने योगदान को निर्धारित करने की अनुमति देता है।

उपर्युक्त कथनों के संबंध में निम्नलिखित में से कौन सा सही है?

a) कथन II और कथन III दोनों सही हैं और ये दोनों कथन I की व्याख्या करते हैं।
b) कथन II और कथन III दोनों सही हैं लेकिन उनमें से केवल एक ही कथन I की व्याख्या करता है।
c) कथन II और III में से केवल एक सही है और यह कथन I की व्याख्या करता है।
d) न तो कथन II और न ही कथन III सही है।

100%		75%		50%		0%	

Q.29) With reference to the Green Credit Programme (GCP) a national initiative by the Ministry of Environment, Forest and Climate Change, Government of India, consider the following statements:

I. The Indian Council of Forestry Research and Education is the Administrator under the Programme.
II. It aims to restore degraded forest lands through the active participation of public and private sector entities.
III. The forest lands restored under the programme can be exchanged for meeting Compensatory Afforestation requirements under the Van (Sanrakshan Evam Samvardhan) adhiniyam, 1980

Which of the statements given above are correct?

a) I and II only
b) I, II and III
c) I and III only
d) II and III only

100%		75%		50%		0%	

Q.30) With reference to Indian Defence sector, consider the following Pairs:

Name of the Missile	Features
I. Pralay Missile	Short range cruise missile
II. STAR Missile	Supersonic target missile
III. Dhvani Missile	Hypersonic glide missile

How many of the above given pairs are correctly matched?

a) Only one
b) Only two
c) All the three
d) None

100%		75%		50%		0%	

Q.31) Which one of the following statements is true with regard to a greenhouse?

a) The shorter wavelength infrared radiations from the Sun can enter into the greenhouse while longer wavelength infrared radiations from the ground and the plants inside the greenhouse cannot pass back out through the glass.
b) The shorter wavelength infrared radiations from the Sun can enter into the greenhouse and the longer wavelength infrared radiations from the ground and the plants inside the greenhouse can also pass back out through the glass.
c) The shorter wavelength infrared radiations from the Sun cannot enter into the greenhouse while longer wavelength infrared radiations from the ground and the plants inside the greenhouse can pass back out through the glass.
d) No infrared radiation can pass through the glass of the greenhouse.

100%		75%		50%		0%	

Q.32) Consider the following international conventions:

I. International Convention for the Prevention of Pollution from Ships, 1973/78 (MARPOL 73/78)
II. Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter, 1972 (London Convention, 1972).
III. International Convention on Maritime Search and Rescue, 1979 (SAR 1979).

How many of the above conventions are ratified by India?

a) Only one
b) Only two
c) All the three
d) None

100%		75%		50%		0%	

Q.33) The Ministry of Environment, Forest and Climate Change has announced a 'National Designated Authority (NDA),' a mandatory requirement under the provisions of the 2015 Paris Agreement. The primary purpose of NDA is to:

a) Prepare Nationally Determined Contribution targets for the next five years.
b) Enable the Carbon Emissions Trading regime.
c) Prepare and oversee emission intensity targets for each sector.
d) Enhancing climate change education, training, public awareness and public participation.

100%		75%		50%		0%	

Q.29) पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय, भारत सरकार की एक राष्ट्रीय पहल ग्रीन क्रेडिट प्रोग्राम (GCP) के संदर्भ में, निम्नलिखित कथनों पर विचार कीजिए:

I. भारतीय वानिकी अनुसंधान और शिक्षा परिषद (ICFRE) इस कार्यक्रम के तहत प्रशासक है।

II. इसका उद्देश्य सार्वजनिक और निजी क्षेत्र की संस्थाओं की सक्रिय भागीदारी के माध्यम से अपक्षयित वन भूमि को बहाल करना है।

III. कार्यक्रम के तहत बहाल की गई वन भूमि का आदान-प्रदान वन (संरक्षण एवं संवर्धन) अधिनियम, 1980 के तहत प्रतिपूरक वनीकरण आवश्यकताओं को पूरा करने के लिए किया जा सकता है।

उपर्युक्त कथनों में से कौन सा सही है?

a) केवल I और II

b) I, II और III

c) केवल I और III

d) केवल II और III

100%		75%		50%		0%	

Q.30) भारतीय रक्षा क्षेत्र के संदर्भ में, निम्नलिखित युग्मों पर विचार कीजिए:

मिसाइल का नाम	**विशेषताएं**
I. प्रलय मिसाइल	कम दूरी की क्रूज मिसाइल
II. स्टार(STAR) मिसाइल	सुपरसोनिक लक्ष्य मिसाइल
III. ध्वनि मिसाइल	हाइपरसोनिक ग्लाइड मिसाइल

उपर्युक्त युग्मों में से कितने सही सुमेलित हैं?

a) केवल एक

b) केवल दो

c) सभी तीन

d) कोई नहीं

100%		75%		50%		0%	

Q.31) ग्रीन हाउस के विषय में निम्नलिखित में से कौन-सा एक कथन सही है ?

a) सूर्य से आने वाला अपेक्षाकृत लघु तरंगदैर्ध्य अवरक्त (इन्फ्रारेड) विकिरण ग्रीन हाउस में प्रवेश कर सकता है, जबकि भूमि और ग्रीन हाउस के अन्दर के पादपों से उत्सर्जित अपेक्षाकृत लंबी तरंगदैर्ध्य अवरक्त (इन्फ्रारेड) विकिरण वापस काँच के पार नहीं जा सकता ।

b) सूर्य से आने वाले अपेक्षाकृत लघु तरंगदैर्ध्य अवरक्त (इन्फ्रारेड) विकिरण ग्रीन हाउस में प्रवेश कर सकता है, और भूमि और ग्रीन हाउस के अन्दर के पादपों से उत्सर्जित अपेक्षाकृत लंबी तरंगदैर्ध्य अवरक्त (इन्फ्रारेड) विकिरण भी वापस काँच के पार जा सकता है।

c) सूर्य से आने वाला अपेक्षाकृत लघु तरंगदैर्ध्य अवरक्त (इन्फ्रारेड) विकिरण ग्रीन हाउस में प्रवेश नहीं कर सकता, जबकि भूमि और ग्रीन हाउस के अन्दर के पादपों से उत्सर्जित अपेक्षाकृत लंबी तरंगदैर्ध्य अवरक्त (इन्फ्रारेड) विकिरण वापस काँच के पार जा सकता है।

d) कोई भी अवरक्त (इन्फ्रारेड) विकिरण ग्रीनहाउस के काँच में से पार नहीं जा सकता ।

100%		75%		50%		0%	

Q.32) निम्नलिखित अंतर्राष्ट्रीय सम्मेलनों पर विचार कीजिए:

I. जहाजों से होने वाले प्रदूषण की रोकथाम के लिए अंतर्राष्ट्रीय सम्मेलन, 1973/78 (MARPOL 73/78)

II. अपशिष्ट और अन्य पदार्थों के डंपिंग द्वारा समुद्री प्रदूषण की रोकथाम पर सम्मेलन, 1972 (लंदन कन्वेंशन, 1972)।

III. समुद्री खोज और बचाव पर अंतर्राष्ट्रीय सम्मेलन, 1979 (SAR 1979)।

उपर्युक्त में से कितने सम्मेलनों को भारत द्वारा अनुमोदित किया गया है?

a) केवल एक

b) केवल दो

c) सभी तीन

d) कोई नहीं

100%		75%		50%		0%	

Q.33) पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय ने 2015 के पेरिस समझौते के प्रावधानों के तहत एक अनिवार्य आवश्यकता 'राष्ट्रीय नामित प्राधिकरण (NDA)' की घोषणा की है। NDA का प्राथमिक उद्देश्य है:

a) अगले पांच वर्षों के लिए राष्ट्रीय स्तर पर निर्धारित योगदान लक्ष्य तैयार करना।

b) कार्बन उत्सर्जन व्यापार व्यवस्था को सक्षम करना।

c) प्रत्येक क्षेत्र के लिए उत्सर्जन तीव्रता लक्ष्य तैयार करना और उनकी देखरेख करना।

d) जलवायु परिवर्तन शिक्षा, प्रशिक्षण, जन जागरूकता और जन भागीदारी को बढ़ाना।

100%		75%		50%		0%	

Q.34) Consider the following statements with respect to the Greenhouse Gas Emission Intensity (GEI) Target Rules, 2025:

I. It set India's first legally binding emission reduction targets for carbon-heavy industries.

II. It covers both public and private carbon emission-intensive sectors.

Which of the statements given above is/are correct?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.35) Consider the following statements:

Statement I: India's pharmaceutical industry has gained international recognition as the "Pharmacy of the World".

Statement II: India ranks 3rd globally in terms of value in pharmaceutical production.

Statement III: India produces more than 50% of the world's vaccines.

Which one of the following is correct in respect of the above statements?

a) Both Statement II and Statement III are correct and both of them explain Statement I
b) Both Statement II and Statement III are correct but only one of them explains Statement I
c) Only one of the Statements II and III is correct and that explains Statement I
d) Neither Statement II nor Statement III is correct

100%		75%		50%		0%	

Q.36) The Integrity Matters Checklist, a tool to help companies and investors align their climate disclosures with the United Nations' standards for credible net-zero commitments and transition plans is launched by:

a) Global Reporting Initiative
b) Transparency International
c) Climate Policy Initiative
d) The Climate Group

100%		75%		50%		0%	

Q.37) Consider the following reports:

I. Global Methane Status Report
II. Emissions Gap Report
III. Global Carbon Budget Report

Which of the above reports are released/ published by the United Nations Environment Programme (UNEP)?

a) I and II only
b) II and III only
c) I and III only
d) I, II and III

100%		75%		50%		0%	

Q.38) Consider the following statements about the Intergovernmental Panel on Climate Change (IPCC) :

I. It is the United Nations body for assessing the science related to climate change.

II. It was set up in 1988 by the World Meteorological Organization (WMO) and United Nations Environment Programme (UNEP)

Which of the statements given above is/are correct?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.39) Consider the following statements:

I. India is the world's largest emitter of nitrous oxide (N_2O).

II. The agriculture sector contributes around 70% to nitrous oxide emissions in India.

Which of the statements given above is/are correct?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.34) ग्रीनहाउस गैस उत्सर्जन तीव्रता (GEI) लक्ष्य नियम, 2025 के संबंध में निम्नलिखित कथनों पर विचार कीजिए:

I. इसने कार्बन-गहन उद्योगों के लिए भारत का पहला कानूनी रूप से बाध्यकारी उत्सर्जन कटौती लक्ष्य निर्धारित किया है।

II. यह सार्वजनिक और निजी दोनों कार्बन उत्सर्जन-गहन क्षेत्रों को शामिल करता है।

उपर्युक्त कथनों में से कौन-सा/से सही है/हैं?

a) केवल I

b) केवल II

c) I और II दोनों

d) न तो I और न ही II

100%		75%		50%		0%	

Q.35) निम्नलिखित कथनों पर विचार कीजिए:

कथन I: भारत के फार्मास्युटिकल (औषधि) उद्योग ने "विश्व की फार्मेसी" के रूप में अंतरराष्ट्रीय मान्यता प्राप्त की है।

कथन II: फार्मास्युटिकल उत्पादन के मूल्य के मामले में भारत वैश्विक स्तर पर तीसरे स्थान पर है।

कथन III: भारत दुनिया के 50% से अधिक टीकों का उत्पादन करता है।

उपर्युक्त कथनों के संबंध में निम्नलिखित में से कौन सा सही है?

a) कथन II और कथन III दोनों सही हैं और ये दोनों कथन I की व्याख्या करते हैं

b) कथन II और कथन III दोनों सही हैं लेकिन उनमें से केवल एक कथन I की व्याख्या करता है

c) कथन II और III में से केवल एक सही है और यह कथन I की व्याख्या करता है

d) न तो कथन II और न ही कथन III सही है।

100%		75%		50%		0%	

Q.36) कंपनियों और निवेशकों को उनके जलवायु प्रकटीकरण को संयुक्त राष्ट्र के विश्वसनीय नेट-ज़ीरो प्रतिबद्धताओं और संक्रमण योजनाओं के मानकों के साथ संरेखित करने में सहायता करने वाला एक उपकरण, 'इंटीग्रिटी मैटर्स चेकलिस्ट' किसके द्वारा लॉन्च किया गया है:

a) ग्लोबल रिपोर्टिंग इनिशिएटिव

b) ट्रांसपेरेंसी इंटरनेशनल

c) क्लाइमेट पॉलिसी इनिशिएटिव

d) द क्लाइमेट ग्रुप

100%		75%		50%		0%	

Q.37) निम्नलिखित रिपोर्टों पर विचार कीजिए:

I. वैश्विक मीथेन स्थिति रिपोर्ट

II. उत्सर्जन अंतराल रिपोर्ट

III. वैश्विक कार्बन बजट रिपोर्ट

उपर्युक्त में से कौन-सी रिपोर्ट संयुक्त राष्ट्र पर्यावरण कार्यक्रम (UNEP) द्वारा जारी/प्रकाशित की जाती हैं?

a) केवल I और II

b) केवल II और III

c) केवल I और III

d) I, II और III

100%		75%		50%		0%	

Q.38) जलवायु परिवर्तन पर अंतर-सरकारी पैनल (IPCC) के बारे में निम्नलिखित कथनों पर विचार कीजिए:

I. यह जलवायु परिवर्तन से संबंधित विज्ञान का आकलन करने के लिए संयुक्त राष्ट्र का निकाय है।

II. इसे 1988 में विश्व मौसम विज्ञान संगठन (WMO) और संयुक्त राष्ट्र पर्यावरण कार्यक्रम (UNEP) द्वारा स्थापित किया गया था।

उपर्युक्त में से कौन-सा/से कथन सही है/हैं?

a) केवल I

b) केवल II

c) I और II दोनों

d) न तो I और न ही II

100%		75%		50%		0%	

Q.39) निम्नलिखित कथनों पर विचार कीजिए:

I. भारत नाइट्रस ऑक्साइड (N_2O) का विश्व का सबसे बड़ा उत्सर्जक है।

II. भारत में नाइट्रस ऑक्साइड उत्सर्जन में कृषि क्षेत्र का योगदान लगभग 70% है।

उपर्युक्त में से कौन-सा/से कथन सही है/हैं?

a) केवल I

b) केवल II

c) I और II दोनों

d) न तो I और न ही II

100%		75%		50%		0%	

Q.40) With reference to the Mines and Minerals (Development and Regulation) Amendment Act, 2025, which one of the following statements is correct?

a) It allows the central government to determine royalty rates and other payments for minor minerals.
b) It establishes an authority for registering and regulating mineral exchanges.
c) It prohibits the National Mineral Exploration and Development Trust from funding mine development activities.
d) It allows inclusion of atomic minerals of any grade into any existing non-atomic mineral lease.

100%		75%		50%		0%	

Q.41) The scientific view is that the increase in global temperature should not exceed 2°C above pre-industrial level. If the global temperature increases beyond 3° C above the pre-industrial level, what can be its possible impact/impacts on the world?

1. Terrestrial biosphere tends toward a net carbon source.
2. Widespread coral mortality will occur.
3. All the global wetlands will permanently disappear.
4. Cultivation of cereals will not be possible anywhere in the world.

Select the correct answer using the code given below:

a) 1 only
b) 1 and 2 only
c) 2, 3 and 4 only
d) 1, 2, 3 and 4

100%		75%		50%		0%	

Q.42) Consider the following pairs:

Terms in News	Description
I. Emissions Gap	It represents the difference between current actions to reduce greenhouse gases (GHGs) and what is needed to meet the target.
II. Carbon Budget	The amount of greenhouse gases that can be emitted for a given level of global warming.
III. Climate Sensitivity	These are markers of a larger climate system which when triggered beyond a threshold, perpetuates warming on its own.

How many of the above pairs are correctly matched?

a) Only one
b) Only two
c) All the three
d) None

100%		75%		50%		0%	

Q.43) Consider the following:

I. Increased amount of Water Vapor in the atmosphere.
II. Ocean Warming
III. Increased Antarctic Rainfall
IV. Loss of Soil Carbon

How many of the above are considered as Positive Climate Feedback Loops i.e. accentuate warming?

a) Only one
b) Only two
c) Only three
d) All the four

100%		75%		50%		0%	

Q.44) Which of the following are the outcomes of/launched at the Conference of the Parties to the United Nations Framework Convention on Climate Change (COP30)?

I. Scaling J-REDD+ Coalition
II. Tropical Forest Forever Facility (TFFF)
III. Climate and Health Funders Coalition (CHFC)
IV. Leadership Group for Industry Transition (LeadIT)

Select the correct answer using the code given below:

a) I, III and IV only
b) I, II and III only
c) II and IV only
d) I, II, III and IV

100%		75%		50%		0%	

Q.40) खान और खनिज (विकास और विनियमन) संशोधन अधिनियम, 2025 के संदर्भ में, निम्नलिखित में से कौन-सा कथन सही है?

a) यह केंद्र सरकार को लघु खनिजों के लिए रॉयल्टी दरों और अन्य भुगतानों को निर्धारित करने की अनुमति देता है।
b) यह खनिज एक्सचेंजों के पंजीकरण और विनियमन के लिए एक प्राधिकरण की स्थापना करता है।
c) यह राष्ट्रीय खनिज अन्वेषण और विकास ट्रस्ट को खनन विकास गतिविधियों के वित्तपोषण से प्रतिबंधित करता है।
d) यह किसी भी श्रेणी के परमाणु खनिजों को किसी भी मौजूदा गैर-परमाणु खनिज पट्टे में शामिल करने की अनुमति देता है।

100%		75%		50%		0%	

Q.41) वैज्ञानिक दृष्टिकोण यह है कि वैश्विक तापमान में वृद्धि पूर्व-औद्योगिक स्तर से 2°C से अधिक नहीं होनी चाहिए। यदि वैश्विक तापमान पूर्व-औद्योगिक स्तर से 3°C से अधिक बढ़ जाता है, तो विश्व पर इसके संभावित प्रभाव क्या हो सकते हैं?

1. स्थलीय जीवमंडल एक शुद्ध कार्बन स्रोत की ओर प्रवृत्त हो जाता है।
2. व्यापक प्रवाल मृत्यु होगी।
3. सभी वैश्विक आर्द्रभूमियाँ स्थायी रूप से समाप्त हो जाएंगी।
4. विश्व में कहीं भी अनाज की खेती संभव नहीं होगी।

नीचे दिए गए कूट का उपयोग करके सही उत्तर चुनिए:

a) केवल 1
b) केवल 1 और 2
c) केवल 2, 3 और 4
d) 1, 2, 3 और 4

100%		75%		50%		0%	

Q.42) निम्नलिखित युग्मों पर विचार कीजिए:

चर्चित शब्द	विवरण
I. एमिशन गैप	यह ग्रीन हाउस गैसों (GHGs) को कम करने के लिए वर्तमान कार्यों और लक्ष्य को प्राप्त करने के लिए आवश्यक कार्यों के बीच के अंतर को दर्शाता है।
II. कार्बन बजट	एक निश्चित स्तर के वैश्विक तापन के लिए उत्सर्जित की जा सकने वाली ग्रीन हाउस गैसों की मात्रा।
III. क्लाइमेट सेंसिटिविटी	ये एक व्यापक जलवायु तंत्र के संकेतक हैं, जो एक बार किसी सीमा (threshold) से अधिक सक्रिय हो जाने पर स्वयं ही तापन की प्रक्रिया को बनाए रखते हैं।

उपर्युक्त में से कितने युग्म सही सुमेलित हैं?

a) केवल एक
b) केवल दो
c) सभी तीन
d) कोई नहीं

100%		75%		50%		0%	

Q.43) निम्नलिखित पर विचार कीजिए:

I. वायुमंडल में जलवाष्प की मात्रा में वृद्धि।
II. महासागरों का ऊष्मीकरण
III. अंटार्कटिका वर्षा में वृद्धि
IV. मृदा कार्बन का ह्रास

उपर्युक्त में से कितने सकारात्मक जलवायु प्रतिपुष्टि चक्र (Positive Climate Feedback Loop) के रूप में माने जाते हैं, अर्थात् तापन को बढ़ाते हैं?

a) केवल एक
b) केवल दो
c) केवल तीन
d) सभी चार

100%		75%		50%		0%	

Q.44) निम्नलिखित में से कौन से संयुक्त राष्ट्र जलवायु परिवर्तन फ्रेमवर्क कन्वेंशन (COP30) के पक्षकारों के सम्मेलन के परिणाम हैं/इस सम्मेलन में शुरू किए गए हैं?

I. स्केलिंग J-REDD+ कोएलिशन
II. ट्रॉपिकल फॉरेस्ट फॉरएवर फैसिलिटी (TFFF)
III. क्लाइमेट एंड हेल्थ फंडर्स कोएलिशन (CHFC)
IV. लीडरशिप ग्रुप फॉर इंडस्ट्री ट्रांजिशन (LeadIT)

नीचे दिए गए कूट का उपयोग करके सही उत्तर चुनिए:

a) केवल I, III और IV
b) केवल I, II और III
c) केवल II और IV
d) I, II, III और IV

100%		75%		50%		0%	

Q.45) In the recent past, the Gola-Tiwai Complex has caught international attention for which one of the following reasons?

a) It is widely considered as the oldest temple complex in the world.
b) A large renewable energy project has been established here
c) The world's largest reserve of gold has been discovered here.
d) It has been designated as the UNESCO World Heritage Site (WHS).

100%		75%		50%		0%	

Q.46) A hiker walks 300 m straight from his camp, turns right and walks 200 m, again turns right and walks 100 m, then turns right once more and walks 400 m to reach a lake. In which direction does he walk initially from his camp if the lake is exactly in the South-West direction?

a) North
b) South
c) East
d) West

100%		75%		50%		0%	

Q.47) A Question is given followed by two Statements I and II. Consider the Question and the Statements.

There are three distinct prime numbers whose sum is an even number.

Question: What are those three numbers?
Statement-I: Their sum is less than 14.
Statement-II: One of the numbers is 3.

Which one of the following is correct in respect of the above Question and the Statements?

a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone
b) The Question can be answered by using either Statement alone
c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone
d) The Question cannot be answered even by using both the Statements together

100%		75%		50%		0%	

Q.48) A Question is given followed by two Statements I and II. Consider the Question and the Statements.

Question: Is (x + y) an integer?
Statement I: (5x + 3y) is an integer.
Statement II: (2x + y) is an integer.

Which one of the following is correct in respect of the above Question and the Statements?

a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone
b) The Question can be answered by using either Statement alone
c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone
d) The Question cannot be answered even by using both the Statements together

100%		75%		50%		0%	

Directions for the following 2 (two) items:
Read the following passage and answer the items that follow the passage. Your answers to these items should be based on the passage only

Passage - 1

The rapid expansion of digital connectivity has transformed online gaming from a niche form of entertainment into a vast and influential digital ecosystem embedded in the daily lives of billions, especially adolescents and young adults. However, this growth has brought with it a range of serious concerns. Real money gaming platforms, betting applications, and unregulated gaming environments have created opportunities for organised criminal networks to exploit users through scams, deceptive practices, and even money laundering. At the same time, online gaming platforms are emerging as potential channels for extremist networking and radicalisation. Alongside these security risks, the psychological and social consequences of excessive gaming, such as addiction, social isolation, and declining mental well being, are increasingly disrupting real world relationships and overall quality of life. As an entire generation grows up immersed in these digital environments, the critical challenge for governments, technology companies, and civil society is to ensure that these platforms do not expose users to financial, psychological, and social harm.

Q.45) हाल के समय में, गोला-तिवाई कॉम्प्लेक्स निम्नलिखित में से किस कारण से अंतरराष्ट्रीय ध्यान में आया है?

a) इसे व्यापक रूप से विश्व का सबसे प्राचीन मंदिर परिसर माना जाता है।
b) यहाँ एक बड़ा नवीकरणीय ऊर्जा परियोजना स्थापित की गई है।
c) यहाँ विश्व का सबसे बड़ा स्वर्ण भंडार खोजा गया है।
d) इसे यूनेस्को विश्व धरोहर स्थल (WHS) के रूप में नामित किया गया है।

100%		75%		50%		0%	

Q.46) एक पदयात्री अपने शिविर से 300 मीटर सीधा चलता है, दाईं ओर मुड़ता है और 200 मीटर चलता है, फिर दाईं ओर मुड़ता है और 100 मीटर चलता है, उसके बाद फिर से दाईं ओर मुड़ता है और 400 मीटर चलकर एक झील तक पहुँचता है। यदि झील ठीक दक्षिण-पश्चिम दिशा में है, तो वह अपने शिविर से प्रारंभ में किस दिशा में चलता है?

a) उत्तर
b) दक्षिण
c) पूर्व
d) पश्चिम

100%		75%		50%		0%	

Q.47) एक प्रश्न दिया गया है, जिसके बाद दो कथन I और II दिए गए हैं। प्रश्न और कथनों पर विचार कीजिए।
तीन भिन्न अभाज्य संख्याएँ हैं जिनका योग एक सम संख्या है।
प्रश्न: वे तीन संख्याएं क्या हैं?
कथन-I: उनका योग 14 से कम है।
कथन-II: संख्याओं में से एक 3 है।
उपर्युक्त प्रश्न और कथनों के संबंध में निम्नलिखित में से कौन-सा सही है?

a) प्रश्न का उत्तर केवल एक कथन से दिया जा सकता है, लेकिन दूसरे कथन से नहीं
b) प्रश्न का उत्तर किसी भी एक कथन से दिया जा सकता है
c) प्रश्न का उत्तर दोनों कथनों को साथ में उपयोग करके दिया जा सकता है, लेकिन किसी एक कथन से नहीं
d) प्रश्न का उत्तर दोनों कथनों को साथ में उपयोग करने पर भी नहीं दिया जा सकता

100%		75%		50%		0%	

Q.48) एक प्रश्न दिया गया है, जिसके बाद दो कथन I और II दिए गए हैं। प्रश्न और कथनों पर विचार कीजिए।
प्रश्न: क्या (x + y) एक पूर्णांक है?
कथन I: (5x + 3y) एक पूर्णांक है।
कथन II: (2x + y) एक पूर्णांक है।
उपर्युक्त प्रश्न और कथनों के संबंध में निम्नलिखित में से कौन-सा सही है?

a) प्रश्न का उत्तर केवल एक कथन से दिया जा सकता है, लेकिन दूसरे कथन से नहीं
b) प्रश्न का उत्तर किसी भी एक कथन से दिया जा सकता है
c) प्रश्न का उत्तर दोनों कथनों को साथ में उपयोग करके दिया जा सकता है, लेकिन किसी एक कथन से नहीं
d) प्रश्न का उत्तर दोनों कथनों को साथ में उपयोग करने पर भी नहीं दिया जा सकता

100%		75%		50%		0%	

निम्नलिखित 2 (दो) प्रश्नों के लिए निर्देश:
निम्नलिखित परिच्छेद को पढ़िए और उसके बाद दिए गए प्रश्नों के उत्तर दीजिए। इन प्रश्नों के उत्तर केवल परिच्छेद के आधार पर होने चाहिए।

परिच्छेद - 1

डिजिटल कनेक्टिविटी के तेजी से विस्तार ने ऑनलाइन गेमिंग को मनोरंजन के एक छोटे रूप से बदलकर एक विशाल और प्रभावशाली डिजिटल इकोसिस्टम में बदल दिया है, जो अरबों लोगों, विशेष रूप से किशोरों और युवाओं के दैनिक जीवन में समाया हुआ है। हालांकि, इस वृद्धि के साथ कई गंभीर चिंताएँ भी उत्पन्न हुई हैं। वास्तविक धन वाले गेमिंग प्लेटफॉर्म, सट्टेबाजी (बेटिंग) एप्लिकेशन और अनियमित गेमिंग वातावरण ने संगठित आपराधिक नेटवर्क के लिए घोटाले, भ्रामक प्रथाओं और यहाँ तक कि मनी लॉन्ड्रिंग के माध्यम से उपयोगकर्ताओं का शोषण करने के अवसर पैदा किए हैं। साथ ही, ऑनलाइन गेमिंग प्लेटफॉर्म चरमपंथी नेटवर्किंग और कट्टरपंथ (रेडिकलाइजेशन) के संभावित माध्यमों के रूप में उभर रहे हैं। इन सुरक्षा जोखिमों के साथ-साथ, अत्यधिक गेमिंग के मनोवैज्ञानिक और सामाजिक परिणाम, जैसे कि लत (एडिक्शन), सामाजिक अलगाव और गिरता मानसिक स्वास्थ्य, वास्तविक दुनिया के रिश्तों और जीवन की समग्र गुणवत्ता को तेजी से बाधित कर रहे हैं। जैसे-जैसे एक पूरी पीढ़ी इन डिजिटल वातावरणों में गहनता से जुड़ी हुई बड़ी हो रही है, सरकारों, प्रौद्योगिकी कंपनियों और नागरिक समाज के लिए यह एक महत्वपूर्ण चुनौती है कि वे सुनिश्चित करें कि ये प्लेटफॉर्म उपयोगकर्ताओं को वित्तीय, मनोवैज्ञानिक और सामाजिक हानि के प्रति उजागर न करें।

Q.49) Which one of the following statements best reflects the most logical and rational inference that can be drawn from the passage?

a) By often doing more harm than good, online gaming platforms negatively impact users' well-being.
b) The expansion of internet connectivity has fueled the growth of online gaming platforms.
c) As online gaming continues to grow, we must ensure it does not harm users' well-being.
d) As children are more susceptible to the negative effects of online gaming, access should be restricted for them.

100%		75%		50%		0%	

Q.50) With reference to the above passage, the following assumptions have been made:

I. Online gaming platforms can be exploited to incite social unrest among communities.
II. Since games involving real money are susceptible to fraud, they should be banned.

Which of the above assumptions is/are valid?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

100%		75%		50%		0%	

Q.49) निम्नलिखित में से कौन-सा कथन परिच्छेद से निकाला जा सकने वाला सबसे तार्किक और युक्तिसंगत निष्कर्ष सर्वोत्तम रूप से व्यक्त करता है?

a) अक्सर लाभ के बजाय अधिक नुकसान पहुँचाकर, ऑनलाइन गेमिंग प्लेटफॉर्म उपयोगकर्ताओं के कल्याण पर नकारात्मक प्रभाव डालते हैं।

b) इंटरनेट कनेक्टिविटी के विस्तार ने ऑनलाइन गेमिंग प्लेटफॉर्म के विकास को बढ़ावा दिया है।

c) जैसे-जैसे ऑनलाइन गेमिंग का विस्तार हो रहा है, हमें यह सुनिश्चित करना चाहिए कि यह उपयोगकर्ताओं के कल्याण को नुकसान न पहुँचाए।

d) चूँकि बच्चे ऑनलाइन गेमिंग के नकारात्मक प्रभावों के प्रति अधिक संवेदनशील होते हैं, इसलिए उनके लिए इसकी पहुंच को प्रतिबंधित किया जाना चाहिए।

100%		75%		50%		0%	

Q.50) उपर्युक्त परिच्छेद के संदर्भ में, निम्नलिखित धारणाएं बनाई गई हैं:

I. ऑनलाइन गेमिंग प्लेटफॉर्म का उपयोग समुदायों के बीच सामाजिक अशांति भड़काने के लिए किया जा सकता है।

II. चूँकि वास्तविक धन से जुड़े खेल धोखाधड़ी के प्रति संवेदनशील होते हैं, इसलिए उन्हें प्रतिबंधित कर देना चाहिए।

उपर्युक्त में से कौन-सी धारणा/धारणाएँ मान्य है/हैं?

a) केवल I

b) केवल II

c) I और II दोनों

d) न तो I और न ही II

100%		75%		50%		0%	

SPACE FOR ROUGH WORK

Q.1) Consider the following statements:

Statement I: Article 6 of the Paris Agreement on climate change is frequently discussed in global discussions on sustainable development and climate change.

Statement II: Article 6 of the Paris Agreement on climate change sets out the principles of carbon markets.

Statement III: Article 6 of the Paris Agreement on climate change intends to promote inter-country non-market strategies to reach their climate targets.

Which one of the following is correct in respect of the above statements?

a) Both Statement II and Statement III are correct and both of them explain Statement I

b) Both Statement II and Statement III are correct but only one of them explains Statement I

c) Only one of the Statements II and III is correct and that explains Statement I

d) Neither Statement II nor Statement III is correct

Ans) a

Exp) Option a is the correct answer.

Statement I is correct: Article 6 of the Paris Agreement is frequently discussed in relation to sustainable development and climate change due to its provisions for international cooperation in the implementation of Nationally Determined Contributions (NDCs). It enables countries to voluntarily cooperate through carbon markets (6.2 and 6.4) and non-market approaches (6.8) to achieve their climate targets, while also promoting sustainable development and environmental integrity.

Statement II is correct: Article 6 of the Paris Agreement on climate change sets out the principles for carbon markets. It provides a framework for international cooperation on emissions reductions, enabling countries to work together through carbon trading and other mechanisms to achieve their climate targets. Article 6 allows countries to transfer emission reductions from one country to another, helping them meet their nationally determined contributions (NDCs).

Statement III is correct: Article 6 of the Paris Agreement facilitates both market-based and non-market-based international cooperation for countries to achieve their climate targets. It's a key mechanism for countries to collaborate in reducing emissions and meeting their Nationally Determined Contributions (NDCs)

- **Article 6.2:** Enables countries to use internationally transferred mitigation outcomes (like carbon credits) towards their NDCs, promoting cooperation and sharing of emissions reduction efforts.
- **Article 6.4:** Establishes a new UNFCCC mechanism for trading high-quality carbon credits, further facilitating international carbon markets.
- **Article 6.8:** Specifically addresses non-market approaches, including finance, technology transfer, and capacity building to enhance climate action. This allows for cooperation without direct trading of emissions reductions, promoting a broader range of collaboration. **Both statement II and III explains statement I**

Source: UPSC CSE Pre 2025

Subject:) Environment

Topic:) Climate Change related initiatives, conventions and organization

Subtopic:)

Q.2) Consider the following International Environmental convention/protocol:

I. Rotterdam Convention

II. Montreal Protocol

III. Stockholm Convention

IV. Basel Convention

With reference to the convention/protocol mentioned above, which one of the following is the correct chronology of the years in which they entered into force?

a) II - IV - III - I

b) II - IV - I -III

c) IV - II - I - III

d) IV - II - III - I

Ans) b

Exp) Option b is the correct answer.

Option II: Montreal Protocol (1989): The Montreal Protocol on Substances that Deplete the Ozone Layer is a global agreement to protect the Earth's ozone layer by phasing out the chemicals that deplete it. This phase-out plan includes both the production and consumption of ozone-depleting substances. **The landmark agreement was signed in 1987 and entered into force in 1989.**

Option IV: Basel Convention (1992) was adopted in 1989. It aims to regulate the **transboundary movement and disposal of hazardous wastes**, aiming to prevent dumping of toxic waste, especially in developing countries. The **Convention entered into force in 1992.**

Option I: The **Rotterdam Convention (February, 2004)** establishes a **Prior Informed Consent (PIC) procedure** for international trade in certain hazardous chemicals and pesticides, ensuring that countries are informed about and consent to the import of such substances.

The text of the Rotterdam Convention was adopted on 10 September 1998 in Rotterdam, the Netherlands. The C**onvention entered into force on 24 February 2004.**

Option III: Stockholm Convention (May, 2004): The Stockholm Convention on Persistent Organic Pollutants is a global treaty to protect human health and the environment from chemicals that remain intact in the environment for long periods, become widely distributed geographically, accumulate in the fatty tissue of humans and wildlife, and have harmful impacts on human health or on the environment. **The Stockholm Convention on Persistent Organic Pollutants was adopted on 22 May 2001 in Stockholm, Sweden. The Convention entered into force on 17 May 2004.**

Hence, Correct chronological order: **II – IV – I - III.**

Source:)

https://www.pic.int/TheConvention/Overview#:~:text=The%20text%20of%20the%20Rotterdam,force%20on%2024%20February%202004.

https://indianexpress.com/article/technology/science/ozone-day-2023-montreal-protocol-8942426/

https://www.basel.int/Implementation/PublicAwareness/20thAnniversary/tabid/1531/Default.aspx

https://www.pops.int/TheConvention/Overview/tabid/3351/Default.aspx

Subject:) Environment

Topic:) Climate Change related initiatives, conventions and organization

Subtopic:)

Q.3) With reference to the Carbon Credit Trading Scheme (CCTS) in India, consider the following statements:

I. Each carbon credit certificate issued will represent a hundred tons of CO_2 equivalent (tCO_2e) reduction (or removal) from the atmosphere.

II. It operates through two mechanisms that is the Compliance Mechanism and the Offset Mechanism.

III. The Central Electricity Regulatory Commission is the agency responsible for issuing carbon credit certificates.

Which of the statements given above is/are correct?

a) I and II only

b) II only

c) II and III only

d) None of the above

Ans) b

Exp) Option b is the correct answer.

The Carbon Credit Trading Scheme (CCTS) has been notified with the objective of reducing, removing, or avoiding greenhouse gas emissions from the Indian economy by pricing such emissions through the trading of Carbon Credit Certificates (CCC).

Statement I is incorrect. **Each Carbon Credit Certificate issued will represent one ton (not hundred) of CO2 equivalent (tCO2e) reduction (or removal) from the atmosphere.**

Statement II is correct. The scheme is designed with a **dual structure**. It includes a **compliance mechanism** for obligated, energy-intensive sectors, and also an **offset mechanism**.

Under the Compliance Mechanism, emission-intensive industries designated as Obligated Entities are required to meet assigned Greenhouse Gas Emission Intensity (GEI) targets, and entities that outperform their targets are eligible for Carbon Credit Certificates.

Under the Offset Mechanism, Non-Obligated Entities may voluntarily register projects that reduce, remove or avoid greenhouse gas emissions for the purpose of seeking issuance of Carbon Credit Certificates.

Statement III is incorrect. **The Bureau of Energy Efficiency (BEE) issues the carbon credits certificate based on the recommendation of the National Steering Committee for the Indian carbon marke**t and subsequent approval of the Central Government. The Central Government constituted the National Steering Committee for Indian Carbon Market (NSCICM) under the Carbon Credit Trading Scheme (CCTS). It will oversee the functioning of the Indian Carbon Market. The committee consists of members from different Ministries under the Chairmanship of Secretary, Ministry of Power and Co-Chairmanship of Secretary, Ministry of Environment, Forest, and Climate Change.

The Grid Controller of India (GCI) is the registry operator and is to manage and operate the Indian Carbon Market registry.

The Central Electricity Regulatory Commission (CERC) is the regulator for trading activities under the Indian Carbon Market.

Source: https://beeindia.gov.in/carbon-market.php#:~:text=To%20achieve%20the%20aforesaid%20objectives,(or%20removal)%20from%20atmosphere.
https://www.pib.gov.in/PressReleasePage.aspx?PRID=2198780&reg=3&lang=2

Subject:) Environment

Topic:) Climate Change related initiatives, conventions and organization

Subtopic:)

Q.4) The "Keeling Curve", a significant indicator in climate science, is associated with the long-term monitoring of which of the following?

a) Glacial retreat in Arctic and Antarctic regions

b) Ocean acidity

c) Atmospheric carbon dioxide (CO_2) concentrations

d) Stratospheric ozone depletion

Ans) c

Exp) Option c is the correct answer.

The **Keeling Curve is a graph** that **represents the concentration of carbon dioxide (CO_2) in Earth's atmosphere over time**, based on continuous measurements taken at the **Mauna Loa Observatory in Hawaii since 1958.**

Knowledge Base:

Keeling Curve:

- **Named after**: Dr. Charles David Keeling, who began the systematic recording of atmospheric CO_2.
- **Shape of the curve**: The graph shows a steadily rising trend in CO_2 levels, with annual seasonal fluctuations.
- **Seasonal pattern**: CO_2 levels drop during the Northern hemisphere's spring and summer due to photosynthesis and rise in fall and winter as vegetation decays.
- **Scientific importance:**
  - Provided the first clear evidence of rising atmospheric CO_2 due to human activities such as fossil fuel combustion and deforestation.
  - It has become one of the most iconic and important datasets in climate science.

Source: https://keelingcurve.ucsd.edu/
https://www.thehindu.com/sci-tech/science/carbon-dioxide-in-the-air-at-highest-level-since-measurements-began/article34766866.ece

Forum IAS Red Book – Environment – Chapter:9- Climate Change

Subject:) Environment

Topic:) Climate Change related initiatives, conventions and organization

Subtopic:)

Q.5) Consider the following statements with respect to the Climate and Clean Air Coalition (CCAC):
I. Its primary objective is to reduce long-lived climate pollutants i.e. those pollutants that remain in the atmosphere for a much longer period of time than carbon dioxide.
II. It mandates binding emission reduction targets for its member countries.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) d
Exp) Option d is the correct answer.
Statement I is incorrect: The Climate and Clean Air Coalition (CCAC) focuses on reducing the super pollutants, which are short-lived in the atmosphere (not long-lived) climate pollutants (SLCPs) such as methane, black carbon, tropospheric ozone and hydrofluorocarbons.
Short-lived climate pollutants are powerful climate forcers that remain in the atmosphere for a much shorter period of time than carbon dioxide, yet their potential to warm the atmosphere can be many times greater. Certain short-lived climate pollutants are also air pollutants that have harmful effects for people, ecosystems and agricultural productivity.
Statement II is incorrect: CCAC does not impose or mandate binding emission reduction targets. Participation in CCAC is voluntary and based on cooperative initiatives rather than legally enforceable obligations. It is a voluntary partnership of governments, intergovernmental organizations, businesses, scientific institutions, and civil society. The UN Environment Programme (UNEP) hosts its Secretariat, but it is not a formal protocol under the UNFCCC.
Source:) https://www.ccacoalition.org/
https://www.ccacoalition.org/content/short-lived-climate-pollutants
Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.6) The Montreal Protocol on Substances that Deplete the Ozone Layer is a global agreement to protect the Earth's ozone layer by phasing out the chemicals that deplete it. In this context, this phase-out plan includes:
I. Production of ozone-depleting substances.
II. Consumption of ozone-depleting substances.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) c
Exp) Option c is the correct answer
Statements I and II are correct: The Montreal Protocol on Substances that Deplete the Ozone Layer is a global agreement to protect the Earth's ozone layer by phasing out the chemicals that deplete it. **This phase-out plan includes both the production and consumption of ozone-depleting substances.** The landmark agreement was signed in 1987 and entered into force in 1989.
Knowledge Base:

- The Kigali Amendment (2016), to the Montreal Protocol mandates a phase-down of Hydrofluorocarbons (HFCs), recognizing their high global warming potential. hydrofluorocarbons (HFCs), were introduced as non-ozone depleting alternatives to support the timely phase-out of CFCs and HCFCs.
- The Montreal Protocol is the first treaty to achieve universal ratification, with 198 parties (including all UN member states and the European Union), reflecting global consensus on ozone protection.

Source: https://ozone.unep.org/treaties/montreal-protocol

Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.7) With reference to the 'Wet-bulb temperature', consider the following statements:
I. It is the lowest temperature of air that can be reached by evaporating water into the air, at constant pressure.
II. High wet-bulb temperatures can limit the human body's ability to cool itself through sweating.
III. Higher humidity leads to a greater difference between dry-bulb and wet-bulb temperatures.
Which of the statements given above are correct?
a) I and II only
b) II and III only
c) I and III only
d) I, II and III

Ans) a
Exp) Option a is the correct answer.
Statement I is correct: Wet-bulb temperature is the lowest temperature of air that can be reached by evaporating water into the air at constant pressure. The wet-bulb temperature is reached when the rate of cooling by evaporation is balanced by the heat gained from the surrounding air.
Statement II is correct: High wet-bulb temperatures reduce the effectiveness of sweat evaporation from the human body. It impairs the body's ability to dissipate heat and can lead to heat stress. If the wet-bulb temperature is high, the air is so saturated with moisture that sweat cannot evaporate effectively.
Statement III is incorrect: Higher humidity means the air is already saturated with water vapor, which slows down evaporation. **As humidity increases, evaporation decreases, and the wet-bulb temperature approaches the dry-bulb temperature, reducing the difference between them, not increasing it.** In 100% humidity, the dry-bulb and wet-bulb temperatures are identical.
The wet bulb temperature is usually lower than the dry bulb temperature, and the difference between the two increases dramatically as the air becomes dry.
Source:) https://www.thehindu.com/opinion/editorial/a-cautionary-tale/article65144683.ece https://forumias.com/blog/wet-bulb-temperature/ https://timesofindia.indiatimes.com/city/hyderabad/wet-bulb-effect-can-prove-fatal-caution-experts/articleshow/120958610.cms
https://www.ndtv.com/science/what-is-wet-bulb-temperature-how-it-works-5755962
Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.8) In the context of climate studies, the term 'Snow Persistence' measures the:
a) fraction of time snow remains on the ground after snowfall.
b) total depth of snow accumulated on a mountain peak during a single winter season.
c) reflectivity of snow-covered surfaces influencing Earth's energy balance.
d) process of transformation of snow into glacial ice over time.

Ans) a
Exp) Option a is the correct answer.
Snow persistence measures the fraction of time snow remains on the ground after snowfall.
It is an important indicator in climate studies for understanding seasonal snow cover, water availability, and climate change impacts.
When this snow melts, it provides water to people and ecosystems. As per the International Centre for Integrated Mountain Development report, snow persistence over the Hindu Kush Himalaya region between November and March 2025 was 23.6 per cent below normal levels, a record low in the last 23 years.
Source:) https://www.newsonair.gov.in/snow-persistence-in-hindu-kush-himalaya-hits-record-low-23-6-below-normal-icimod-report/

https://www.thehindu.com/sci-tech/energy-and-environment/hindu-kush-himalayas-snow-update-2024-findings/article68307604.ece
Subject:) Environment
Topic:) Mitigation Strategies
Subtopic:)

Q.9) With reference to Ocean Darkening, consider the following statements:
I. It is the shrinking of the photic zones or those layers of water where sunlight can pass and induce the process of photosynthesis
II. Increased sediment runoff from rivers and coastal erosion contributes to ocean darkening.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) c
Exp) Option c is the correct answer.
Statement I is correct: Ocean darkening is the decline in the transparency of seawater, leading to a reduction in the depth and intensity of sunlight penetration. This phenomenon shrinks the photic zone, which is critical for photosynthesis. This directly affects marine ecosystems and primary productivity.
Statement II is correct: Increased sediment runoff, coastal erosion, dissolved organic matter, and pollutants enhance turbidity and light absorption, contributing to ocean darkening. Dissolved Organic Matter (DOM) and suspended sediments runoff from rivers absorbs UV and blue light before it can reach the depths of ocean water.
Source:) https://indianexpress.com/article/technology/science/global-oceans-darkening-disrupting-life-beneath-surface-10036814/
https://www.weforum.org/videos/ocean-darkening/
Subject:) Environment
Topic:) Ocean acidification and ozone depletion
Subtopic:)

Q.10) In the context of the financial sector, which of the following best describes the term 'Co-lending':
a) A system in which two borrowers jointly apply for a single loan to reduce their individual credit burden.
b) A credit arrangement in which two banks compete to offer a lower interest rate to the same borrower.
c) A financing arrangement where a bank and a Non-Banking Financial Company (NBFC) jointly provide a loan and share funds and risks.
d) A process where a bank refinances loans already issued by an NBFC to improve the NBFC's liquidity.

Ans) c
Exp) Option c is the correct answer.
Co-lending refers to a structured **lending arrangement** in which a **bank and a Non-Banking Financial Company** (NBFC) **jointly extend a loan to the same borrower**.
Under this model, both institutions contribute funds in a predetermined ratio, share the associated credit risks, and sometimes the expertise. This enables more efficient credit delivery, improves access to underserved segments, and allows lenders to expand their customer base while maintaining balanced risk exposure.
The **RBI** has recently **issued revised guidelines** to **strengthen the co-lending framework** between banks and the NBFCs.
Source: SFG Essential, Current Affairs, Quarterly Compilation (July–September 2025), Page: 58-59.
Subject:) Current Affairs
Topic:) Economy
Subtopic:)

Q.11) Which one of the following statements best describes the term 'Social Cost of Carbon'?
It is a measure, in monetary value, of the
a) long-term damage done by a tonne of CO_2 emissions in a given year.
b) requirement of fossil fuels for a country to provide goods and services to its citizens, based on the burning of those fuels.
c) efforts put in by a climate refugee to adapt to live in a new place.
d) contribution of an individual person to the carbon footprint on the planet Earth.

Ans) a
Exp) Option a is the correct answer
The Social Cost of Carbon is usually estimated as the net present value of climate change impacts over the next 100 years (or longer) of one additional tonne of carbon emitted to the atmosphere today. It is the marginal global damage costs of carbon emissions.
The social cost of carbon (SCC) is the central concept for the inclusion of climate change damages in the Cost-Benefit Analysis of public policy and public investments. It measures the present value in monetary terms of the damages incurred when an additional ton of carbon (or any other Greenhouse gas) is released into the atmosphere.
Source: UPSC CSE Pre 2020
Subject:) Environment
Topic:) Mitigation Strategies
Subtopic:)

Q.12) In the context of the Paris Agreement, consider the following statements about the Biennial Transparency Report (BTR):
I. Only the developed countries that are parties to the Paris agreement are required to submit the report every two years.
II. It is part of the Enhanced Transparency framework of the Paris Agreement and replaces the earlier 'Biennial Update Reports' and 'Biennial Reports' submitted by countries.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) b
Exp) Option b is the correct answer.
The Biennial Transparency Report (BTR) is a key component of the Enhanced Transparency Framework (ETF) under the Paris Agreement, designed to foster international trust and track global climate efforts.
Statement I is incorrect. The biennial transparency report (BTR) under the Paris Agreement is a transparency-oriented report that is to be produced by **participating countries (both developed and developing)** every two years and submitted to the UNFCCC. It includes information on national inventory reports (NIR), progress towards Nationally Determined Contributions (NDCs), policies and measures, climate change impacts and adaptation, levels of financial, technology development and transfer and capacity-building support, capacity-building needs and areas of improvement. Small Islands Developing States (SIDS) and Least Developed Countries (LDCs) may submit the information required for the BTR at their discretion.
Statement II is correct. The Paris Agreement's Enhanced Transparency Framework (ETF) for Action and Support aims to build mutual trust between Parties under the United Nations Framework Convention on Climate Change (UNFCCC). **The Biennial Transparency Report (BTR) is part of the Enhanced Transparency framework, which replaces both the Biennial Report and the Biennial Update Report under the UNFCCC.**
Source:) https://www.thehindu.com/sci-tech/energy-and-environment/india-to-submit-updated-greenhouse-gas-inventory-to-un-soon/article69234598.ece
https://www.thehindubusinessline.com/news/india-china-miss-deadline-for-first-biennial-transparency-report/article69048985.ece

Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.13) The triple planetary crisis refers to the three main interlinked issues that humanity currently faces. It includes:
a) climate change, biodiversity loss, and pollution.
b) climate change, economic recession, and political instability.
c) biodiversity loss, desertification, and food insecurity.
d) pollution, global conflict, and migration.

Ans) a
Exp) Option a is the correct answer.
The term "Triple Planetary Crisis" describes the three main interlinked environmental challenges that humanity is currently facing: climate change, pollution, and biodiversity loss. These three crises are not isolated; they feed into one another and must be addressed collectively to ensure a sustainable future.
Source:)
https://unfccc.int/news/what-is-the-triple-planetary-crisis
Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.14) It is one of the widest glaciers in the world, spanning approximately 120 km. Often referred to as the 'Doomsday Glacier,' its rapid melting is a major concern for global sea-level rise, as its collapse could eventually trigger the loss of a significant portion of the ice sheet.
Which of the following glaciers is being described above?
a) Lambert Glacier
b) Yala Glacier
c) Thwaites Glacier
d) Jakobshavn Glacier

Ans) c
Exp) Option c is the correct answer.
The glacier described is the **Thwaites Glacier, located in West Antarctica**. **Spanning roughly 120 km,** it is recognized as the widest glacier on Earth. It is famously referred to as the "**Doomsday Glacier**" because it acts as a critical natural dam, holding back a massive portion of the West Antarctic Ice Sheet. Its rapid melting is a major concern for scientists because its collapse could eventually trigger a catastrophic rise in global sea levels, potentially by several feet, which would threaten coastal regions worldwide.
Source:) https://www.thehindu.com/sci-tech/science/why-does-thwaites-glacier-matter/article70607073.ece
https://indianexpress.com/article/explained/explained-thwaites-glacier-finding-doomsday-melting-worry-7269124/
Subject:) Environment
Topic:) Climate Change and its Impact
Subtopic:)

Q.15) Consider the following:
I. A significant rise in the temperature in the regions of North Western Europe
II. Alter the timing of Indian Monsoons
III. A rise in sea levels along the North American Atlantic coast
IV. A global increase in the rate of ocean carbon sequestration
How many of the above are possible consequences of the weakening of Atlantic Meridional Overturning Circulation (AMOC)?
a) Only one

b) Only two
c) Only three
d) All the four

Ans) b

Exp) Option b is the correct answer.

The Atlantic Meridional Overturning Circulation (AMOC) is a massive system of ocean currents that acts as a global "conveyor belt," transporting warm surface water from the tropics toward the North Atlantic. Its weakening or collapse would have profound global impacts.

Statement I is incorrect. Gulf Stream, a part of the AMOC, is a warm current responsible for the mild climate on the Eastern coast of North America as well as Europe. **Without a proper AMOC and Gulf Stream, Europe will be very cold. Thus, it would not lead to rise in temperature of north western Europe rather Europe will be cold.**

Statement II is correct. The AMOC influences the position of the Intertropical Convergence Zone (ITCZ). A slowdown would likely shift tropical rainfall belts southward, **disrupting monsoon cycles in West Africa and India.**

Statement III is correct. The motion of AMOC currents helps pull water away from the U.S. East Coast. **If the current slows down, water "piles up" along the North American Atlantic coast.**

Statement IV is incorrect. The AMOC plays a vital role in transporting dissolved carbon dioxide from the surface to the deep ocean. **A weakening of this system would decrease (not increase) the ocean's ability to sequester carbon.**

Source:) https://indianexpress.com/article/explained/explained-climate/collapse-atlantic-current-10220675/
https://www.thehindu.com/sci-tech/energy-and-environment/the-gulf-stream-will-not-collapse-in-2025-what-the-alarmist-headlines-got-wrong/article67161497.ece

Subject:) Environment
Topic:) Climate Change and its Impact
Subtopic:)

Q.16) With reference to hydrogen production, hydrogen produced through electrolysis of water using electricity generated from nuclear energy is known as:

a) Pink Hydrogen
b) Turquoise Hydrogen
c) White Hydrogen
d) Black Hydrogen

Ans) a

Exp) Option a is the correct answer

Hydrogen is seen as a clean energy carrier in the global transition away from fossil fuels.

Depending on the source of energy and production process of hydrogen, the carbon footprint varies significantly. To simplify this variation, a color-coded terminology has been developed.

The "color" assigned to hydrogen depends on three key factors:

- Feedstock used: natural gas, coal, water, biomass
- Production process: electrolysis, gasification, reforming
- Carbon emissions and their management: whether CO_2 is released, captured, or avoided

Colour						
Type	Black/Brown Hydrogen	Grey Hydrogen	Blue Hydrogen	Turquise Hydrogen	Pink Hydrogen	Green Hydrogen
Process	Coal Gasification	Methane Reformation	Coal Gasification &Methane Reformation with CCUS	Pyrolysis	Electrolysis	Electrolysis/ Biomass Gasification
Source	Coal	Natural Gas	Fossil Fuel	Methane	Nuclear Energy	Renewable Energy

Pink (or purple) hydrogen is **produced through electrolysis of water using electricity generated from nuclear energy** and is considered a low-carbon hydrogen source.
Knowledge Base:
Types of Hydrogen

- **Blue hydrogen** is produced from fossil fuels similar to grey hydrogen, but **it incorporates carbon capture and storage technology** to reduce carbon dioxide emissions.
- **Green hydrogen** is produced by the **electrolysis of water** using electricity generated from renewable energy sources such as solar, wind, or hydropower, and it does not produce carbon emissions.
- **Brown hydrogen** is produced through the **gasification of lignite** coal, leading to very high carbon dioxide emissions.
- **White hydrogen** refers to naturally occurring hydrogen found in underground geological formations, which can be extracted without industrial production processes.
- **Turquoise hydrogen** is produced through **methane pyrolysis**, where natural gas is split into hydrogen and **solid carbon (instead of CO_2)**.
- **Black hydrogen** is generated by the **gasification of bituminous coal** and is associated with high levels of carbon emissions.
- **Grey hydrogen** is produced from natural gas through steam methane reforming, and it results in significant carbon dioxide emissions without any carbon capture.

Source:) https://www.weforum.org/stories/2021/07/clean-energy-green-hydrogen/
https://static.pib.gov.in/WriteReadData/specificdocs/documents/2024/may/doc2024510336301.pdf (Page 17)
Subject:) Environment
Topic:) Mitigation Strategies
Subtopic:)

Q.17) With reference to the ecological impacts of marine heat waves, consider the following statements:
I. It may promote the growth of harmful algal blooms in coastal waters.
II. It may cause large-scale dieback of seagrass meadows.
III. It can alter the migration pattern of marine species.
IV. It will boost the growth of Kelp forests.
Which of the statements given above is/are correct?
a) I and II only
b) I, II and III only
c) III and IV only
d) I, II, III and IV

Ans) b
Exp) Option b is the correct answer.
A marine heat wave is an extreme weather event. It occurs when the surface temperature of a particular region of the sea rises to 3 or 4 degree Celsius above the average temperature for at least five days. It can last for weeks, months or even years.
Statement I is correct: Marine heat waves may promote the occurrence of harmful algal blooms in coastal waters. Warmer sea temperatures create **favourable growth conditions for phytoplankton**, especially toxin-producing species. Reduced vertical mixing during heat waves allows these algae to **accumulate near the surface**, leading to blooms.
statement II is correct: Marine heat waves may cause large-scale dieback of seagrass meadows and kelp forests. Seagrasses are sensitive to **temperature stress**, and prolonged warming disrupts their **photosynthesis and respiration balance**. Higher temperatures also reduce dissolved oxygen, leading to **hypoxic conditions**.
Statement III is correct: Marine heat waves may cause the deaths of several marine species, alter their migration patterns. For example, in 2012, Marine Heat Wave over the northwest Atlantic Ocean caused marine species that like warm water to move northwards and migrate earlier than they usually did, affecting fisheries targeting those species.

statement IV is incorrect: Marine heat waves may destroy kelp forests and fundamentally altered the ecosystem of the coast. Kleps usually grow in cooler waters, providing habitat and food for many marine animals.
Source:) https://egyankosh.ac.in/bitstream/123456789/83548/1/Unit-10.pdf
https://www.thehindu.com/news/national/marine-heatwaves-rising-around-india-says-study/article38393806.ece
Subject:) Environment
Topic:) Climate Change and its Impact
Subtopic:)

Q.18) Which of the following is/are causes of coral bleaching?
I. Decreased solar radiation
II. Ocean alkalinization
III. Extreme low tides
IV. Nutrient enrichment of oceanic water
Select the correct answer using the code given below:
a) I and II only
b) IV only
c) III and IV only
d) II, III and IV

Ans) c
Exp) Option c is the correct answer.
Most corals contain algae called zooxanthellae – they are plant-like organisms – in their tissues. Corals and zooxanthellae have a symbiotic relationship. While corals provide zooxanthellae a safe place to live, zooxanthellae provide oxygen and organic products of photosynthesis that help corals to grow and thrive. Zooxanthellae give bright and unique colours to corals. Corals are very sensitive to light and temperature and even a small change in their living conditions can stress them. When stressed, they expel zooxanthellae and turn entirely white. This is called coral bleaching.
Coral bleaching doesn't immediately lead to the death of corals. They rather go under more stress and are subject to mortality. Coral bleaching reduces the reproductivity of corals and makes them more vulnerable to fatal diseases. If the bleaching is not too severe, corals have been known to recover.
Option I is incorrect: Coral bleaching is primarily triggered by stress factors such as elevated sea surface temperatures and high solar radiation, particularly ultraviolet (UV) rays. **Decreased solar radiation does not cause bleaching; rather, reduced light intensity can sometimes help alleviate stress on corals.**
Option II is incorrect: The relevant phenomenon affecting corals is **ocean acidification**, not alkalinization. Ocean acidification occurs due to increased absorption of atmospheric CO_2, which lowers seawater pH and reduces the availability of carbonate ions **necessary** for coral skeleton formation. While this weakens coral health, **alkalinization** is being explored as a **mitigation strategy** rather than a cause of coral bleaching.
Option III is correct: Extreme low tides: Exposure to the air during extreme low tides can cause bleaching in shallow corals.
Option IV is correct: Nutrient enrichment: Increased nutrient concentrations affect corals by promoting phytoplankton growth, which in turn supports increased numbers of organisms that compete with coral for space.
Source:) https://forumias.com/blog/fifth-mass-coral-bleaching-event/
https://indianexpress.com/article/explained/explained-climate/global-mass-coral-bleaching-triggered-9279731/
https://www.thehindu.com/sci-tech/energy-and-environment/global-coral-bleaching-event-expands-now-the-largest-on-record/article68767949.ece https://www.downtoearth.org.in/climate-change/caribbean-coral-reefs-have-halved-since-1980s-as-heat-disease-intensify https://www.livemint.com/opinion/online-views/climate-catastrophe-bleached-coral-should-rattle-the-world-into-action-11715267419350.html
Subject:) Environment
Topic:) Ocean acidification and ozone depletion
Subtopic:)

Q.19) With reference to environmental governance, which one of the following is related to the concept of 'GreenRinsing'?
a) A practice where firms rely on collective industry commitments to substitute for individual sustainability action.
b) A practice where firms are adopting and later abandoning ambitious net-zero targets to secure investments.
c) A practice where environmental organisations collectively lobby corporations to adopt net-zero pledges.
d) A practice where companies overstate environmental benefits using sector-wide aggregate data.

Ans) b
Exp) Option b is the correct answer.
Firms are adopting and later abandoning ambitious net-zero targets to secure investments, a practice termed as "greenrinsing". This reflects a trend where companies set unrealistic climate goals but fail to achieve them. The company regularly changing its Environmental Social Governance targets before they are achieved.
Knowledge Base:

- **Greenlighting** occurs when company communications (including advertisements) spotlight a particularly green feature of its operations or products, however small, in order to draw attention away from environmentally damaging activities being conducted elsewhere.
- **Greenshifting** is when companies imply that the consumer is at fault and shift the blame on to them.
- **Greenlabelling** is a practice where marketers call something green or sustainable, but a closer examination reveals this to be misleading.
- **Greencrowding** is a practice in which companies rely on collective industry actions or group affiliations to create an impression of environmental responsibility while avoiding meaningful individual sustainability efforts.
- **Greenhushing** refers to corporate management teams under-reporting or hiding their sustainability credentials in order to evade investor scrutiny.

Source:)
https://initiatives.weforum.org/sustainable-finance-data-skills-and-capacity-building/knowledge-hub/publications/'greenrinsing':-firms-setting-and-dropping-net-zero-targets-to-create-revolving-door-for-finance,-says-un-official/64a98c1d95559ded4dd6d2210a2c41b3b467da08
https://planet-tracker.org/greenwashing-growing-increasingly-sophisticated-says-planet-tracker/
https://planet-tracker.org/wp-content/uploads/2023/01/Greenwashing-Hydra-3.pdf
Subject:) Environment
Topic:) Mitigation Strategies
Subtopic:)

Q.20) Consider the following statements:
I. India is the world's top producer of aluminium globally.
II. Rajasthan holds the highest share of copper reserves in India.
III. Chile has the largest known copper reserves in the world.
Which of the statements given above are correct?
a) I and II only
b) I and III only
c) II and III only
d) I, II and III

Ans) c
Exp) Option c is the correct answer.
Statement I is incorrect: China dominates global aluminium production, accounting for around 58% of the world's total output. Other major producers include Australia, Brazil, and India, but **India does not occupy the top position globally.**
In **India**, the **largest bauxite reserves** are found in **Odisha**, followed by Chhattisgarh and Andhra Pradesh. **Odisha** also **leads in bauxite production**, contributing the highest share to the country's total output.

Statement II is correct: Rajasthan possesses 52.25% of India's copper reserves, the **largest among all states**. Other copper-bearing states include Madhya Pradesh and Jharkhand.

Statement III is correct: Chile is known as the **copper capital of the world** owing to its massive reserves and dominant production share. **Chile holds 19% of global copper reserves**, the **highest among all countries** in the world. It is followed by countries such as Peru and Australia, each holding around 10%.

Source: SFG Essential, Current Affairs, Quarterly Compilation (July–September 2025), Page: 60-61

Subject:) Current Affairs

Topic:) Economy

Subtopic:)

Q.21) Consider the following statements:

Statement I: Scientific studies suggest that a shift is taking place in the Earth's rotation and axis.

Statement II: Solar flares and associated coronal mass ejections bombarded the Earth's outermost atmosphere with a tremendous amount of energy.

Statement III: As the Earth's polar ice melts, the water tends to move towards the equator.

Which one of the following is correct in respect of the above statements?

a) Both Statement II and Statement III are correct and both of them explain Statement I

b) Both Statement II and Statement III are correct but only one of them explains Statement I

c) Only one of the Statements II and III is correct and that explains Statement I

d) Neither Statement II nor Statement III is correct

Ans) b

Exp) Option b is the correct answer.

Statement I is correct: Scientific studies confirm that the Earth's rotation and axis are experiencing shifts, primarily influenced by climate change and human activities. These changes include a wobble in the axis of rotation and a slowing of the Earth's spin.

Statement II is correct: Solar flares and coronal mass ejections (CMEs) are powerful bursts of energy and matter released from the sun. They can bombard Earth's atmosphere with energy and most of this energy is reflected back into space by the Earth's magnetic field. However, these energy bursts **do not contribute to shifts in Earth's rotation or axis.**

Statement III is correct: When polar ice caps and glaciers melt, the resulting water redistributes across the planet, primarily moving towards the equator due to gravitational equilibrium and centrifugal forces from Earth's rotation. This redistribution of mass has a measurable effect on the inertia and balance of the planet, which can lead to a shift in the rotation axis. This is much like how a spinning top shifts direction if its internal weight is redistributed.

Statement III is correct and the **primary explanation** for why the Earth's axis is shifting – due to mass redistribution from melting ice moving toward the equator. Therefore, the correct answer is b.

Source: UPSC CSE Pre 2025

Subject:) Environment

Topic:) Climate Change and its Impact

Subtopic:)

Q.22) With reference to the Aerosols, consider the following statements:

I. Sulfate and nitrate aerosols have a net cooling effect on the climate by reflecting incoming solar radiation back into space.

II. Black carbon aerosols near the surface absorb heat and warm the surrounding air

III. Rapidly reducing aerosol pollution without simultaneous reduction in greenhouse gases can lead to an increase in atmospheric temperature

Which of the statements given above are correct?

a) I and II only

b) II and III only

c) I and III only

d) I, II and III

Ans) d

Exp) Option d is the correct answer.

Aerosols are tiny solid or liquid particles suspended in the atmosphere. They originate from natural sources like dust, sea salt, volcanic ash as well as anthropogenic sources like industrial emissions, biomass burning.

Statement I is correct: Sulfate and nitrate aerosols are scattering aerosols that reflect incoming solar radiation back to space, leading to a net cooling effect on the climate. They also act as Cloud Condensation Nuclei (CCN). By increasing the number of cloud droplets, they make clouds brighter and more reflective.

Statement II is correct: Black carbon aerosols absorb solar radiation and heat the surrounding air, contributing to atmospheric warming. Aerosols like sulfates reflect sunlight and cool the Earth.

Statement III is correct: Aerosols like sulphates have a cooling effect by reflecting sunlight. A rapid reduction in aerosols without reducing greenhouse gases removes this cooling effect, potentially leading to a short-term increase in atmospheric temperature. If aerosol pollution is cleaned to improve public health without simultaneously reducing CO2 and methane, the "mask" by aerosols is removed. Sunlight will hit the Earth with full force, leading to a sudden and accelerated spike in atmospheric temperature.

Source:) https://www.downtoearth.org.in/climate-change/indias-cooling-paradox
https://www.pib.gov.in/PressReleasePage.aspx?PRID=1985770&reg=3&lang=2
https://www.weforum.org/stories/2023/11/air-pollution-aerosols-accelerate-climate-change/
https://www.ccacoalition.org/short-lived-climate-pollutants/black-carbon

Subject:) Environment

Topic:) Climate Change and its Impact

Subtopic:)

Q.23) Which of the following factors primarily contribute to the occurrence of 'thirst waves'?

I. Rapidly rising temperatures.

II. Consistently low atmospheric humidity.

III. Increased wind speed.

IV. High soil moisture levels.

Select the correct answer using the code given below:

a) I, II and IV only

b) III and IV only

c) I, II and III only

d) I, II, III and IV

Ans) c

Exp) Option c is the correct answer.

"Thirst waves" refer to sudden, extreme surges in Atmospheric Evaporative Demand (AED)–essentially a measure of how "thirsty" the atmosphere is for water.

"Thirst wave" to **denote three or more contiguous days of intense evaporative demand**. While heat waves are caused by particular temperature and wind patterns, a thirstwave is the product of temperature, humidity, solar radiation, and wind speed. When temperatures rise, the consequences include more heat as well as the mechanics of water exchange between land and atmosphere, which in turn alters humidity, wind, and solar radiation.

Statement I is correct. Rapidly rising temperatures increase the air's capacity to hold water, which directly boosts the demand for evaporation from the land and vegetation.

Statement II is correct. Consistently low atmospheric humidity (dry air) **creates a larger vapor pressure deficit,** meaning the air can absorb significantly more moisture from the surface.

Statement III is correct. Increased wind speeds help to accelerate evaporation by physically moving moisture-saturated air away from the surface and replacing it with drier air, maintaining a high rate of water loss.

Statement IV is incorrect. High soil moisture levels do not cause the "thirst" of the atmosphere; rather, the atmospheric "thirst wave" occurs regardless of ground conditions and acts to deplete the existing soil moisture, often leading to flash droughts.

Source:) https://www.thehindu.com/sci-tech/energy-and-environment/rising-evaporative-demand-spotlights-indias-data-and-research-gap/article69728191.ece
Subject:) Environment
Topic:) Climate Change and its Impact
Subtopic:)

Q.24) Which of the following best describes the concept of Hydro-Climate Whiplash?
It refers to:
a) the rapid transition between two weather extremes, extremely wet weather and dangerously dry conditions.
b) the gradual long-term decline in average annual precipitation across semi-arid regions.
c) the phenomenon where increased glacial melt causes seasonal flooding and river channel displacement.
d) the intensification of monsoon variability across tropical regions resulting from sea surface temperature anomalies.

Ans) a
Exp) Option a is the correct answer
Hydro-climate whiplash refers to sharp and rapid transition between opposite hydrological extremes–such as intense rainfall/floods followed by sudden droughts (or vice versa).
Hydroclimate whiplash occurs when periods of heavy rainfall lead to rapid vegetation growth, which is subsequently followed by intense heat and drought that dry out the vegetation.
It increases the risk of wildfires because the vegetation grown during wet phases becomes highly flammable fuel during subsequent dry phases. Further it is intensified by climate change, as rising temperatures increase both atmospheric moisture (leading to heavy rainfall) and evaporation (leading to severe droughts). It contributes to multiple environmental hazards, including floods, droughts, wildfires, and ecosystem stress, due to the rapid alternation of hydrological extremes.
Example: January's, 2025 wildfires in Los Angeles were among the worst in California's history. Experts say their ferocity has been fuelled in part by 'hydroclimate whiplash', a climate change-driven phenomenon that is growing in intensity. The **term describes the swing between two weather extremes, extremely wet weather and dangerously dry conditions.**
Source:) https://www.downtoearth.org.in/climate-change/what-is-hydroclimate-whiplash-and-how-did-it-fuel-the-los-angeles-wildfires
https://www.weforum.org/videos/how-hydroclimate-whiplash-is-increasing-the-risk-of-wildfires/
Subject:) Environment
Topic:) Mitigation Strategies
Subtopic:)

Q.25) What is the Rio+20 Conference, often mentioned in the news?
a) It is the United Nations Conference on Sustainable Development
b) It is a Ministerial Meeting of the World Trade Organization
c) It is a Conference of the Inter Government Panel on Climate Change.
d) It is a Conference of the Member Countries of the Convention on Biological Diversity.

Ans) a
Exp) Option a is the correct answer.
The **United Nations Conference on Sustainable Development or Rio+20, took place in Rio de Janeiro, Brazil on 20-22 June 2012.** It resulted in a focused political outcome document which contains clear and practical measures for implementing sustainable development. In Rio, Member States decided to launch a process to develop a set of Sustainable Development Goals (SDGs), which will build upon the Millennium Development Goals and converge with the post 2015 development agenda.
Source: UPSC CSE Pre 2015
Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.26) With reference to the 'Belém 4x Pledge', consider the following statements:
I. It aims to accelerate the clean energy transition through the adoption of clean energy sources, such as biogases and biofuels, as well as hydrogen and synthetic fuels.
II. The pledge is endorsed and supported by India.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) c
Exp) Option c is the correct answer
Statement I is correct: Belém Commitment for Sustainable Fuels (also known as the Belém 4x Pledge), a multilateral initiative launched at COP30 to quadruple the production and use of sustainable fuels by 2035 (base level is 2024). **The Pledge aims to accelerate the clean energy transition through the adoption of clean energy sources, such as biogases and biofuels, as well as hydrogen and synthetic fuels.**
Statement II is correct: The Belém 4x pledge to quadruple sustainable fuel production and use by 2035, put forward by Italy, Japan, India and Brazil and presently supported by 23 countries. **Thus, the pledge was endorsed and supported by India.**
Source: https://www.downtoearth.org.in/climate-change/cop30-implementation-push-across-energy-industry-and-transport-in-belem-as-countries-rally-behind-system-wide-transition
https://www.fao.org/in-action/global-bioenergy-partnership/news-and-events/events/events-detail/the-bel%C3%A9m-4x-pledge--the-contribution-of-sustainable-bioenergy/en
https://www.weforum.org/stories/2025/11/quadrupling-clean-fuels-by-2035/
Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.27) With reference to the Systematic Observation Impact Bond, recently seen in the news, consider the following statements:
I. It is the world's first impact bond aimed at strengthening weather and climate data systems globally.
II. It has been launched by the World Bank.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) a
Exp) Option a is the correct answer.
The Systematic Observation Impact Bond is designed for reliable climate and weather **data and financing gaps through an innovative financial instrument**.
Statement I is correct. The bond is described as the **world's first impact bond focused specifically on weather and climate data**. Its purpose is to improve **systematic observation systems**, especially in Least Developed Countries (LDCs) and Small Island Developing States (SIDS). Better data leads to **better forecasts, stronger early warning systems, and improved climate resilience**.
Statement II is incorrect. The bond is **not launched by the World Bank**. It is being developed under the **Systematic Observations Financing Facility (SOFF)**, which is a United Nations initiative. It is jointly supported by the **United Nations Environment Programme (UNEP), the World Meteorological Organization (WMO), and the United Nations Development Programme (UNDP).**

Source: https://climatepromise.undp.org/news-and-stories/worlds-first-impact-bond-weather-and-climate-data-opens-contributions
https://wmo.int/news/media-centre/worlds-first-impact-bond-weather-and-climate-data-opens-contributions
Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.28) With reference to the Paris Agreement under the United Nations Framework of Climate Change (UNFCCC), consider the following statements:
Statement I: The Paris Agreement seeks to find a Goldilocks solution between the contrasting models of the Kyoto Protocol and the 2009 Copenhagen Accord.
Statement II: The Paris Agreement is a treaty within the meaning of international law, but not all its provisions establish legal obligations.
Statement III: The Paris Agreement takes a bottom-up approach as it allows parties to nationally determine their contributions to address climate change.
Which one of the following is correct in respect of the above statements?
a) Both Statement II and Statement III are correct and both of them explain Statement I.
b) Both Statement II and Statement III are correct but only one of them explains Statement I.
c) Only one of the Statements II and III is correct and that explains Statement I.
d) Neither Statement II nor Statement III is correct.

Ans) a
Exp) Option a is the correct answer
Statement I is correct: The Paris Agreement seeks to find a middle ground – a "Goldilocks" solution – between the contrasting models of the Kyoto Protocol and the 2009 Copenhagen Accord.
Statement II is correct: The Paris Agreement is a treaty within the meaning of international law, but not all its provisions establish legal obligations. Parties to Paris Agreement do not have an obligation to achieve their nationally determined contributions (NDCs) to address climate change – thus, in that respect, NDCs are not legally binding.
Statement III is correct: The Paris Agreement takes a bottom-up approach to the substance of climate change policy, allowing parties to nationally determine their contributions to address climate change.
Statement II explains Statement I: Like the Kyoto Protocol and unlike the Copenhagen Accord, the Paris Agreement is a treaty within the meaning of international law. Thus there is a middle ground or Goldilocks solution in the Paris Agreement.
Statement III explains Statement I: Like the Copenhagen Accord, the Paris Agreement takes a bottom-up approach and in contrast to the Kyoto Protocol, which prescribed emissions limitation targets from the top down, through international negotiations. **Thus there is a middle ground or Goldilocks solution in the Paris Agreement.**
Source: https://legal.un.org/avl/ha/pa/pa.html
Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.29) With reference to the Green Credit Programme (GCP) a national initiative by the Ministry of Environment, Forest and Climate Change, Government of India, consider the following statements:
I. The Indian Council of Forestry Research and Education is the Administrator under the Programme.
II. It aims to restore degraded forest lands through the active participation of public and private sector entities.
III. The forest lands restored under the programme can be exchanged for meeting Compensatory Afforestation requirements under the Van (Sanrakshan Evam Samvardhan) adhiniyam, 1980
Which of the statements given above are correct?
a) I and II only
b) I, II and III
c) I and III only
d) II and III only

Ans) b

Exp) Option b is the correct answer.

The Green Credit Programme (GCP) is a national initiative by the Ministry of Environment, Forest and Climate Change (MoEFCC) to promote environment friendly actions by awarding Green Credits for verified positive environmental activities.

Statement I is correct: The Indian Council of Forestry Research and Education, Dehradun is the Administrator under the Green Credit Programme and is responsible for the effective implementation of the Green Credit Programme, including its management, operation and issuance of green credits under these rules.

Statement II is correct: The Green Credit Programme aims to restore degraded forest lands through the active participation of public and private sector entities. The degraded forest and parcels proposed to be taken up under the Green Credit Programme are elected and registered by the State Forest Department after due verification on the ground.

Statement III is correct: The forest lands restored under Green Credit Programme can be exchanged for meeting Compensatory Afforestation requirements under the Van (Sanrakshan Evam Samvardhan) adhiniyam, 1980

Knowledge Base:

Other Facts about the Green Credit Programme (GCP):

- The Green credits can be claimed by an applicant only after minimum five years of restoration work have been completed and a minimum canopy density of forty per cent has been achieved.
- The Green credits issued for the tree plantation under the GCP shall be non-transferable except between the holding company and its subsidiary companies. The Green Credits issued for tree plantation under GCP shall also be non-tradable.

Source:) https://www.moefcc-gcp.in/faq
https://www.pib.gov.in/PressReleasePage.aspx?PRID=2240640&reg=3&lang=2#:~:text=The%20Green%20Credit%20Programme%20is,been%20developed%20for%20the%20purpose.

Subject:) Environment

Topic:) Climate Change related initiatives, conventions and organization

Subtopic:)

Q.30) With reference to Indian Defence sector, consider the following Pairs:

Name of the Missile	Features
I. Pralay Missile	Short range cruise missile
II. STAR Missile	Supersonic target missile
III. Dhvani Missile	Hypersonic glide missile

How many of the above given pairs are correctly matched?

a) Only one
b) Only two
c) All the three
d) None

Ans) b

Exp) Option b is the correct answer.

Pair I is incorrect: Pralay is an indigenously developed surface-to-surface, short-range quasi-ballistic missile powered by solid propellant and capable of achieving hypersonic speeds above Mach 5. Unlike conventional ballistic missiles, its quasi-ballistic design allows it to follow a lower trajectory and execute mid-flight manoeuvres, enhancing its ability to evade interception. Developed by DRDO's Research Centre Imarat in Hyderabad with support from other DRDO laboratories, Pralay has an operational range of 150–500 km.

Pair II is correct: The STAR Missile is an indigenous high-speed target missile developed by the Defence Research and Development Organisation (DRDO) to provide realistic threat simulation for training the Indian armed forces; it is not meant for combat use. **Designed to mimic enemy missiles for interception practice**, it features a two-stage propulsion system consisting of a solid booster rocket for rapid lift-off and a Liquid Fuel Ramjet (LFRJ) for **sustained supersonic flight.**

Pair III is correct: The Dhvani Missile is a next-generation Hypersonic Glide Vehicle (HGV) being developed by the Defence Research and Development Organisation (DRDO), designed to be launched to extreme altitudes before gliding at speeds exceeding Mach 5 with high manoeuvrability. Expected to be deployed by 2029–30 following successful trials, it offers a remarkable range of 6,000–10,000 km, nearly twice that of Agni-V. The system incorporates advanced scramjet propulsion and thermal shielding technologies derived from the Hypersonic Technology Demonstrator Vehicle (HSTDV), along with ultra-high-temperature ceramic composites capable of withstanding 2,000–3,000°C.
Source: SFG Essential- Current Affairs- Quarterly Compilation (July–September 2025)- Page no: 49-50
Subject:) Current Affairs
Topic:) Science & Technology
Subtopic:)

Q.31) Which one of the following statements is true with regard to a greenhouse?
a) The shorter wavelength infrared radiations from the Sun can enter into the greenhouse while longer wavelength infrared radiations from the ground and the plants inside the greenhouse cannot pass back out through the glass.
b) The shorter wavelength infrared radiations from the Sun can enter into the greenhouse and the longer wavelength infrared radiations from the ground and the plants inside the greenhouse can also pass back out through the glass.
c) The shorter wavelength infrared radiations from the Sun cannot enter into the greenhouse while longer wavelength infrared radiations from the ground and the plants inside the greenhouse can pass back out through the glass.
d) No infrared radiation can pass through the glass of the greenhouse.

Ans) a
Exp) Option a is the correct answer.
A greenhouse is designed to create a favorable environment for plants to thrive in **colder climates.** When sunlight enters the greenhouse, it is absorbed by the plants and surfaces inside. However, much of the outgoing infrared radiation, which carries heat, cannot pass through the glass and becomes trapped inside. This trapped heat creates a warm and conducive environment for plant growth. Similarly, the **Earth can be thought of as a greenhouse.** It **absorbs solar radiation,** particularly in the vi**sible spectrum**, from the Sun. Greenhouse gases, such as **carbon dioxide and methane**, then **retain some of the infrared radiation** emitted by the Earth, preventing it from escaping back into space. The **glass used in a greenhouse is opaque to infrared radiation**, **reflecting and absorbing these longer wavelength** radiations. This property of greenhouse gases and glass helps to **trap the energy from the Sun**, keeping it inside the greenhouse or the Earth's atmosphere, leading to warming effects.
Source: UPSC CDS [I] 2022
Subject:) Environment
Topic:) Climate Change and its Impact
Subtopic:)

Q.32) Consider the following international conventions:
I. International Convention for the Prevention of Pollution from Ships, 1973/78 (MARPOL 73/78)
II. Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter, 1972 (London Convention, 1972).
III. International Convention on Maritime Search and Rescue, 1979 (SAR 1979).
How many of the above conventions are ratified by India?
a) Only one
b) Only two
c) All the three
d) None

Ans) b
Exp) Option b is the correct answer.
Option I is correct: The **International Convention for the Prevention of Pollution from Ships (MARPOL)** is the primary global treaty designed to minimize pollution of the marine environment from ships, whether caused by

operational activities or accidental spills. **India has ratified the MARPOL Convention** (International Convention for the Prevention of Pollution from Ships). In 2018, the Union Cabinet also gave its approval for the ratification of Annex VI of the MARPOL Convention, which specifically deals with the prevention of air pollution from ships.

Option II is incorrect: The London Convention 1972 is one of the **first global conventions** to protect the **marine environment from human activities** and has been in force since 1975. Its **main objective is to promote the effective control of all sources of marine pollution** and to take all practicable steps to prevent pollution of the sea by dumping of wastes and other matter.

India has not yet ratified the London Convention, 1972 (Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter).

Option III is correct: International Convention on Maritime Search and Rescue, 1979, is the primary global legal instrument aimed at developing an international SAR plan, ensuring that no matter where an accident occurs, the rescue of persons in distress at sea will be coordinated by a SAR organization. **India ratified the International Convention on Maritime Search and Rescue, 1979 (SAR 1979) in May 2001.** The Indian Coast Guard serves as the nodal agency for implementing the provisions of this convention in the Indian Search and Rescue Region (ISRR).

Source:)
https://dgshipping.gov.in/Content/PageUrl.aspx?page_name=ShipManualChap10#:~:text=Convention%20on%20the%20Prevention%20of,as%20amended%20(LDC%20(Amended)%201972)&text=India%20has%20ratified%20some%20of,Although%20a%20number%20of%20the
Search And Rescue | Official Website of Indian Coast Guard Head Quarters, New Delhi, Ministry of Defence, Government of India.
https://www.thehindu.com/news/national/cabinet-nod-to-international-pact-on-air-pollution-from-ships/article2281539.ece
Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.33) The Ministry of Environment, Forest and Climate Change has announced a 'National Designated Authority (NDA),' a mandatory requirement under the provisions of the 2015 Paris Agreement. The primary purpose of NDA is to:
a) Prepare Nationally Determined Contribution targets for the next five years.
b) Enable the Carbon Emissions Trading regime.
c) Prepare and oversee emission intensity targets for each sector.
d) Enhancing climate change education, training, public awareness and public participation.

Ans) b
Exp) Option b is the correct answer.
The Ministry of Environment, Forest and Climate Change established the 'National Designated Authority (NDA)' to enable a carbon emissions trading regime.

Option b is correct: The establishment of the **National Designated Authority (NDA)** is a **mandatory requirement under Article 6 of the Paris Agreement (2015)** under the United Nations Framework Convention on Climate Change (UNFCCC). The main objective of the NDA is **to enable the carbon emissions trading regime or carbon markets in India.**

Knowledge Base:
Composition of NDA: It is a 21-member committee headed by the Secretary of the Environment Ministry. Representatives include officials from the Ministry of External Affairs, Ministry of Renewable Energy, NITI Aayog and Steel Ministry among others.

Source:) https://www.thehindu.com/news/national/environment-ministry-finalises-national-designated-authority-to-kickstart-carbon-markets/article69979886.ece
Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.34) Consider the following statements with respect to the Greenhouse Gas Emission Intensity (GEI) Target Rules, 2025:
I. It set India's first legally binding emission reduction targets for carbon-heavy industries.
II. It covers both public and private carbon emission-intensive sectors.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) c
Exp) Option c is the correct answer.
The Ministry of Environment, Forest and Climate Change has notified the Greenhouse Gases Emission Intensity Target Rules, 2025.
Statement I is correct: The Greenhouse Gases Emission Intensity Target Rules, 2025 **sets India's first legally binding emission reduction targets for carbon-heavy industries.** It makes it mandatory for as many as 208 industrial units spread across the country to reduce GHG emissions per unit of product (emission intensity), beginning 2025-26. This includes high-emission sectors: **aluminium, cement, chlor-alkali, pulp & paper, petroleum refinery, petrochemicals, textiles and secondary aluminium.**
Statement II is correct: Greenhouse Gas Emission Intensity (GEI) Target Rules, 2025 **cover both public-sector enterprises (PSUs) and private-sector companies** within designated carbon-intensive sectors. Each facility must reduce the amount of greenhouse gases emitted per unit of output (measured in tonnes of carbon dioxide equivalent per tonne of product) compared to a 2023-24 baseline. The compliance period begins in 2025-26 and continues through 2026-27.
Source:) https://www.thehindu.com/sci-tech/energy-and-environment/india-notifies-first-emission-intensity-targets-for-carbon-intensive-sectors/article70147013.ece
https://timesofindia.indiatimes.com/india/govt-brings-4-more-sectors-under-ghg-emissions-cut-regime/articleshowprint/126709560.cms?val=3728
Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.35) Consider the following statements:
Statement I: India's pharmaceutical industry has gained international recognition as the "Pharmacy of the World".
Statement II: India ranks 3rd globally in terms of value in pharmaceutical production.
Statement III: India produces more than 50% of the world's vaccines.
Which one of the following is correct in respect of the above statements?
a) Both Statement II and Statement III are correct and both of them explain Statement I
b) Both Statement II and Statement III are correct but only one of them explains Statement I
c) Only one of the Statements II and III is correct and that explains Statement I
d) Neither Statement II nor Statement III is correct

Ans) c
Exp) Option c is the correct answer.
Statement I is correct: India has earned the title "Pharmacy of the World" because of its exceptional contribution to global health care.
Key reasons include:

- India is **one of the largest producers and exporters of generic medicines**, supplying low-cost, high-quality drugs to over 200 countries.
- India has a **significant global presence in vaccine manufacturing**, playing a central role in immunization programmes worldwide.

- Its pharmaceutical exports support health systems in both developing and developed nations. India is among the largest exporters of pharmaceutical products, holding a 5.71% share of the global market.

Also, India's pharmaceutical industry is projected to reach USD 130 billion by 2030 and USD 450 billion by 2047.

Statement II is incorrect: India does not rank 3rd in the world in terms of value of pharmaceutical production. India actually **ranks 3rd in volume**, meaning it produces a large quantity of medicines.

But in value terms, India ranks much lower (14th globally), because Indian pharma primarily produces low-cost generics rather than high-value patented drugs.

Statement III is correct: India is the **world's largest producer of vaccines** and contributes roughly **60% of global vaccine output**. The **Serum Institute of India** (SII) is the **largest vaccine manufacturer in the world** by the number of doses produced. This significant contribution strengthens India's role in ensuring global vaccine supply, especially during public health emergencies, as seen during the COVID-19 pandemic.

Conclusion:

- **Statement II is incorrect.**
- **Statement III is correct**, and it **directly supports Statement I** because India's massive vaccine production capacity is one of the key reasons why the world refers to India as the **"Pharmacy of the World."**

Therefore, the **correct answer is: c) Only one of the Statements II and III is correct and that explains Statement I.**

Source: SFG Essential, Current Affairs, Quarterly Compilation (July–September 2025), Page: 64

Subject:) Current Affairs

Topic:) Economy

Subtopic:)

Q.36) The Integrity Matters Checklist, a tool to help companies and investors align their climate disclosures with the United Nations' standards for credible net-zero commitments and transition plans is launched by:

a) Global Reporting Initiative

b) Transparency International

c) Climate Policy Initiative

d) The Climate Group

Ans) a

Exp) Option a is the correct answer.

The **Integrity Matters Checklist** provides a framework for reporting on climate action by non-state actors – including businesses and investors – operationalizing UN guidance on alignment with science-based net zero pathways. This includes how organizations should report on efforts to cut greenhouse gas emissions, implement transition plans, and reduce investment in fossil fuels.

It is launched by the **Global Reporting Initiative (GRI) in partnership with the United Nations.** Its objective is to provide science-based guidance for credible, transparent, and accountable corporate climate commitments.

It aims to bridge the gap between corporate climate ambition and transparency, ensuring that promises match actions. Further, it would strengthen private-sector contributions to both the Paris Agreement and the 2030 Agenda for Sustainable Development.

Source:) https://www.downtoearth.org.in/climate-change/gri-launches-un-endorsed-tool-to-boost-corporate-climate-accountability

https://www.globalreporting.org/news/news-center/gri-launches-new-tool-for-corporate-climate-action/

Subject:) Environment

Topic:) Climate Change related initiatives, conventions and organization

Subtopic:)

Q.37) Consider the following reports:

I. Global Methane Status Report

II. Emissions Gap Report

III. Global Carbon Budget Report

Which of the above reports are released/published by the United Nations Environment Programme (UNEP)?

a) I and II only

b) II and III only
c) I and III only
d) I, II and III

Ans) a
Exp) Option a is the correct answer.
Option I is correct: Global Methane Status Report released by the UN Environment Programme (**UNEP**) and the Climate and Clean Air Coalition (**CCAC**). Assesses gaps in meeting the Global Methane Pledge, which aims to cut methane emissions significantly this decade.
Option II is correct: The Emissions Gap Report is published by the UN Environment Programme (**UNEP**). Emissions Gap Report, 2025 finds that global warming projections over this century, based on full implementation of Nationally Determined Contributions (NDCs), are now 2.3-2.5°C, while those based on current policies are 2.8°C. This compares to 2.6-2.8°C and 3.1°C in last year's report.
Option III is incorrect: The Global Carbon Budget Report was published by the **Global Carbon Project** during UNFCCC's COP29 summit.
Source:)
https://www.thehindu.com/news/cities/mumbai/un-report-flags-indias-rising-methane-emissions-highlights-agricultural-gaps/article70298356.ece
https://www.unep.org/resources/emissions-gap-report-2025
https://www.unep.org/resources/report/global-methane-status-report-2025
https://globalcarbonbudget.org/
Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.38) Consider the following statements about the Intergovernmental Panel on Climate Change (IPCC) :
I. It is the United Nations body for assessing the science related to climate change.
II. It was set up in 1988 by the World Meteorological Organization (WMO) and United Nations Environment Programme (UNEP)
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) c
Exp) Option c is the correct answer
Statement I is correct: The Intergovernmental Panel on Climate Change (IPCC) is the United Nations body for assessing the science related to climate change.
Statement II is correct: The IPCC was set up in 1988 by the World Meteorological Organization (WMO) and United Nations Environment Programme (UNEP) to provide policymakers with regular assessments of the scientific basis of climate change, its impacts and future risks, and options for adaptation and mitigation.
Knowledge Base:

- The assessments by IPCC are policy-relevant but not policy prescriptive: they may present projections of future climate change based on different scenarios and the risks that climate change poses and discuss the implications of response options, **but they do not tell policymakers what actions to take.**
- The IPCC does not conduct its own research.

Source: https://www.ipcc.ch/
https://www.ipcc.ch/site/assets/uploads/2024/04/IPCCFactSheet_WhatisIPCC.pdf
Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.39) Consider the following statements:
I. India is the world's largest emitter of nitrous oxide (N_2O).
II. The agriculture sector contributes around 70% to nitrous oxide emissions in India.
Which of the statements given above is/are correct?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) b
Exp) Option b is the correct answer
Statement I is incorrect: As per India's fourth Biennial Update Report (BUR) submitted to the United Nations Framework Convention on Climate Change (UNFCCC) Nitrous oxide (N_2O) contributes 5.1 percent to India's total GHG emissions. **India is the fourth largest emitter of N_2O after China, the United States of America, and Brazil.**
But as per the latest report, India is the second-largest emitter of nitrous oxide.
Statement II is correct: The agriculture sector contributes 70% to nitrous oxide emissions, followed by energy (16 percent), waste (12 percent) and Industrial Process & Product Use (IPPU) (2 percent). The major sources of N_2O emission from agriculture sector are agricultural soils (89 percent), mainly due to the application of synthetic fertilizers, and manure management (9 percent) and the remaining 2 percent from agricultural residue burning.
Source: https://sansad.in/getFile/annex/267/AU2307_e8pnUh.pdf?source=pqars
https://www.thehindu.com/sci-tech/energy-and-environment/india-is-worlds-second-largest-emitter-of-nitrous-oxide/article68281521.ece
Subject:) Environment
Topic:) Climate Change and its Impact
Subtopic:)

Q.40) With reference to the Mines and Minerals (Development and Regulation) Amendment Act, 2025, which one of the following statements is correct?
a) It allows the central government to determine royalty rates and other payments for minor minerals.
b) It establishes an authority for registering and regulating mineral exchanges.
c) It prohibits the National Mineral Exploration and Development Trust from funding mine development activities.
d) It allows inclusion of atomic minerals of any grade into any existing non-atomic mineral lease.

Ans) b
Exp) Option b is the correct answer.
The **Mines and Minerals (Development and Regulation) Amendment Act, 2025** seeks to liberalize, modernize, and make India's mineral sector more dynamic, with a strong focus on critical and strategic minerals essential for emerging technologies. The Bill introduces significant reforms to the MMDR Act, 1957, aligning the legal framework with contemporary economic, industrial, and technological needs.
Option a is incorrect: It **empowers state governments and not Central Government) to determine the royalty rates** and other payments for **minor minerals** (e.g., building stones, gravel, sand). However, for major minerals included in minor mineral leases, the conditions will be prescribed by the central government.
Option b is correct: It **provides for the establishment** of an **authority responsible for registering and regulating mineral exchanges**, which will function as electronic platforms for trading minerals and metals, bringing transparency and market efficiency.
Option c is incorrect: The **National Mineral Exploration Trust** (NMET) is **renamed the National Mineral Exploration and Development Trust.** Its **mandate is expanded beyond exploration** to also **include funding mine development activities**, thereby accelerating resource utilisation.
Additionally, the contribution to the Trust has been increased from 2% to 3% of royalty.

Option d is incorrect: It**explicitly prohibits** the i**nclusion of atomic minerals above a specified grade**, such as uranium or thorium, **into any mining lease** that was **originally granted for non-atomic minerals**. This safeguard ensures that atomic minerals remain under strict central control.
Source: SFG Essential, Current Affairs, Quarterly Compilation (July–September 2025), Page: 62-63
Subject:) Current Affairs
Topic:) Polity
Subtopic:)

Q.41) The scientific view is that the increase in global temperature should not exceed 2°C above pre-industrial level. If the global temperature increases beyond 3° C above the pre-industrial level, what can be its possible impact/impacts on the world?
1. Terrestrial biosphere tends toward a net carbon source.
2. Widespread coral mortality will occur.
3. All the global wetlands will permanently disappear.
4. Cultivation of cereals will not be possible anywhere in the world.
Select the correct answer using the code given below:
a) 1 only
b) 1 and 2 only
c) 2, 3 and 4 only
d) 1, 2, 3 and 4

Ans) b
Exp) Option b is the correct answer
Options 1 and 2 are correct: If the global temperature increases beyond 3oC above the preindustrial level the terrestrial biosphere which plays an important role in regulating atmospheric composition would tend towards a net carbon source. Also, with increase in temperature mass coral bleaching would occur leading to widespread coral mortality.
Options 3 and 4 are incorrect: All the global wetlands will permanently disappear and cultivation of cereals will not be possible anywhere in the world would be very extreme statements. As temperate regions can still survive with increase in temperature.
Source: UPSC CSE Pre 2014
Subject:) Environment
Topic:) Climate Change and its Impact
Subtopic:)

Q.42) Consider the following pairs:

Terms in News	Description
I. Emissions Gap	It represents the difference between current actions to reduce greenhouse gases (GHGs) and what is needed to meet the target.
II. Carbon Budget	The amount of greenhouse gases that can be emitted for a given level of global warming.
III. Climate Sensitivity	These are markers of a larger climate system which when triggered beyond a threshold, perpetuates warming on its own.

How many of the above pairs are correctly matched?
a) Only one
b) Only two
c) All the three
d) None

Ans) b
Exp) Option b is the correct answer.

Pair I is correctly matched. The Emissions Gap represents the **difference between the greenhouse gas emission levels that countries have committed to achieve through their pledges and the actual reductions** required to limit global warming to a specific target, such as 1.5°C.

Pair II is correctly matched. A Carbon Budget is defined as the total cumulative **amount of carbon dioxide that can be emitted globally while still maintaining a reasonable chance of keeping the Earth's temperature rise below a specific limit.**

Pair III is incorrectly matched. Climate Sensitivity is a measure of how much the Earth's average surface temperature is expected to rise in response to a doubling of atmospheric carbon dioxide concentrations.

Climate tipping points or CTPs are markers of a larger climate system which when triggered beyond a threshold, perpetuate warming on its own.

Source:) https://www.thehindu.com/sci-tech/energy-and-environment/un-emissions-gap-report-3-degrees-warming-2100/article67556832.ece

https://www.thehindu.com/sci-tech/energy-and-environment/cop-28-fair-share-global-carbon-budget-explained/article67564460.ece

https://www.thehindu.com/opinion/op-ed/Climate-tipping-points/article15165692.ece

https://www.thehindu.com/sci-tech/energy-and-environment/studying-earths-distant-past-can-teach-us-lessons-about-its-climate-for-the-future/article5726295.ece

Subject:) Environment

Topic:) Mitigation Strategies

Subtopic:)

Q.43) Consider the following:

I. Increased amount of Water Vapor in the atmosphere.

II. Ocean Warming

III. Increased Antarctic Rainfall

IV. Loss of Soil Carbon

How many of the above are considered as Positive Climate Feedback Loops i.e. accentuate warming?

a) Only one

b) Only two

c) Only three

d) All the four

Ans) d

Exp) Option d is the correct answer.

A Positive Climate Feedback Loop refers to a self-reinforcing cycle where an initial change (like warming) triggers a process that further intensifies that change, accelerating global warming.

Option I is correct. As the atmosphere warms, its capacity to hold water vapor increases. Since **water vapor is a potent greenhouse gas, it traps more heat, leading to further warming** and even more evaporation.

Option II is correct: Ocean warming is an example of a potential positive feedback mechanism. The oceans are an important sink for CO_2 through absorption of the gas into the water surface. As CO_2 increases, it increases the warming potential of the atmosphere. If air temperatures warm, it should warm the oceans. The ability of the ocean to remove CO_2 from the atmosphere decreases with increasing temperature. Because of this, increasing CO_2 in the atmosphere could have effects that actually intensify the increase in CO_2 in the atmosphere.

Option III is correct. Increased Antarctic rainfall accelerates melting. Liquid rain melts snow and ice much faster than warm air does, **revealing darker surfaces that absorb more solar radiation.**

Option IV is correct. Rising temperatures lead to soil carbon loss (especially from thawing permafrost) and more frequent wildfires. **Both processes release massive amounts of stored carbon dioxide and methane into the atmosphere, further driving the greenhouse effect.**

Source:) https://www.thehindu.com/sci-tech/energy-and-environment/feedback-loops-worsening-climate-crisis/article66524620.ece

https://indianexpress.com/article/india/dalai-lama-greta-thunberg-climate-change-discussion-7140973/

https://timesofindia.indiatimes.com/home/environment/global-warming/27-feedback-loops-or-global-warming-accelerators-add-urgency-to-respond-to-climate-crisis-study/articleshow/98046026.cms and https://www.pib.gov.in/PressReleasePage.aspx?PRID=2212322&reg=3&lang=1
Subject:) Environment
Topic:) Climate Change and its Impact
Subtopic:)

Q.44) Which of the following are the outcomes of/launched at the Conference of the Parties to the United Nations Framework Convention on Climate Change (COP30)?
I. Scaling J-REDD+ Coalition
II. Tropical Forest Forever Facility (TFFF)
III. Climate and Health Funders Coalition (CHFC)
IV. Leadership Group for Industry Transition (LeadIT)
Select the correct answer using the code given below:
a) I, III and IV only
b) I, II and III only
c) II and IV only
d) I, II, III and IV

Ans) b
Exp) Option b is the correct answer.
The 30th edition of the Conference of the Parties to the United Nations Framework Convention on Climate Change (UNFCCC COP30) concluded in Belém, Brazil in 2025.
Option I is correct: At COP30 of UNFCCC, partners announced the launch of a new Scaling J-REDD+ (Reducing Emissions from Deforestation and Forest Degradation) Coalition. It is a new multi-stakeholder effort to advance high-integrity jurisdictional REDD+ and support countries seeking to bring jurisdictional credits to market. It is a collective action-orientated effort to accelerate action and finance for forests at the scale needed to meet global climate goals.
Option II is correct: Tropical Forest Forever Facility (TFFF), launched during COP30 of UNFCCC, is a Brazil-led global fund to reward tropical countries for protecting and expanding forests. It aims to mobilise around USD 125 billion through public and private investment, using returns to pay nations that conserve forests.
Option III is correct: The Climate and Health Funders Coalition (CHFC) is a philanthropic alliance announced at COP30 of the UNFCCC. It committed $300 million to address climate change and its health impacts through integrated solutions.
Option IV is incorrect: The Leadership Group for Industry Transition (LeadIT) was launched by the Governments of India and Sweden at the UN Secretary General's Climate Action Summit in 2019 (and not at COP30 of UNFCCC) and is supported by the World Economic Forum (WEF).
Source:) https://www.thehindu.com/sci-tech/energy-and-environment/cop30-summit-india-support-effective-climate-policies/article70313504.ece
https://www.forestcarbonpartnership.org/results-story-announcement/scaling-j-redd-coalition-launched-cop30
https://forestclimateleaders.org/news-and-resources/scaling-j-redd_coalition/
https://tfff.earth/
https://indianexpress.com/article/explained/explained-climate/new-fund-tropical-forests-tfff-10357258/
https://www.hindustantimes.com/india-news/35-philanthropies-vow-300mn-to-help-tackle-impact-of-climate-crisis-101763061309258.html
https://www.pib.gov.in/PressReleasePage.aspx?PRID=1876186&reg=3&lang=2
https://www.pib.gov.in/PressReleseDetail.aspx?PRID=2193076&reg=6&lang=1
Subject:) Environment
Topic:) Climate Change related initiatives, conventions and organization
Subtopic:)

Q.45) In the recent past, the Gola-Tiwai Complex has caught international attention for which one of the following reasons?

a) It is widely considered as the oldest temple complex in the world.
b) A large renewable energy project has been established here
c) The world's largest reserve of gold has been discovered here.
d) It has been designated as the UNESCO World Heritage Site (WHS).

Ans) d
Exp) Option d is the correct answer.
Sierra Leone's **Gola-Tiwai Complex has been designated as the country's first UNESCO World Heritage Site (WHS).** It is a lush forest that is home to one of the world's highest concentrations of primates. Gola is the largest expanse of tropical rainforest in Sierra Leone.
The Complex includes the Gola Rainforest National Park and the Tiwai Island Wildlife Sanctuary.
Source: SFG Essential, Current Affairs, Quarterly Compilation (April–June 2025), Page: 36
Subject:) Current Affairs
Topic:) Places in News
Subtopic:)

Q.46) A hiker walks 300 m straight from his camp, turns right and walks 200 m, again turns right and walks 100 m, then turns right once more and walks 400 m to reach a lake. In which direction does he walk initially from his camp if the lake is exactly in the South-West direction?
a) North
b) South
c) East
d) West

Ans) d
Exp) Option d is the correct answer.
Assume initially the hiker walks **North** and place the camp at **(0, 0)**.
Trace the movement
- Walk **300 m North** → position **(0, 300)**
- Turn right, walk **200 m East** → position **(200, 300)**
- Turn right, walk **100 m South** → position **(200, 200)**
- Turn right, walk **400 m West** → position **(-200, 200)**

Find the relative direction
The final point is **(-200, 200)**.
That means:
- **200 m West**
- **200 m North**

So under the assumption that the initial direction was **North**, the lake lies in the **North-West** direction.
The question says the lake is actually in the **South-West** direction.
So the assumed map must be rotated until **North-West** becomes **South-West**.
That requires a **90° counter-clockwise rotation**.
If the assumed initial direction **North** is rotated **90° counter-clockwise**, it becomes: **West**
Subject:) CSAT
Topic:) Logical and Verbal Reasoning
Subtopic:)

Q.47) A Question is given followed by two Statements I and II. Consider the Question and the Statements.
There are three distinct prime numbers whose sum is an even number.
Question: What are those three numbers?
Statement-I: Their sum is less than 14.
Statement-II: One of the numbers is 3.

Which one of the following is correct in respect of the above Question and the Statements?

a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone

b) The Question can be answered by using either Statement alone

c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

d) The Question cannot be answered even by using both the Statements together

Ans) d

Exp) Option d is the correct answer.

Given there are **three distinct prime numbers** whose sum is an **even number**.

Now:

1. All prime numbers except **2** are odd.
2. Sum of three odd numbers is **odd**.
3. Therefore, for the sum of three distinct primes to be **even**, one of them must be **2**.

So the three numbers must be of the form:

2, p, q where **p** and **q** are distinct odd primes.

Check Statement I alone

Their sum is **less than 14**.

Since one prime must be **2**, we need:

2 + p + q < 14

Now try distinct odd primes:

- **2 + 3 + 5 = 10**
- **2 + 3 + 7 = 12**
- **2 + 5 + 7 = 14** → not allowed

So possible triplets are:

(2, 3, 5) and **(2, 3, 7)**

Thus, **Statement I alone is not sufficient**.

Check Statement II alone

One of the numbers is **3**.

Since the sum is even, one of the primes must be **2**.

So the numbers must be:

2, 3, p where **p** is another distinct prime.

Possible examples:

- **2, 3, 5**
- **2, 3, 7**
- **2, 3, 11**

So **Statement II alone is not sufficient**.

Check both statements together

From Statement II, one number is **3**.

From the parity condition, one number must be **2**.

So the three numbers are: **2, 3, p**

Now use Statement I: **2 + 3 + p < 14**

So: **p < 9**

The only prime numbers less than 9 are: **2, 3, 5, 7**

But the primes must be **distinct**, and **2** and **3** are already used.

So the third number can be: **5** or **7**

Thus we still get two possible triplets:

i. **(2, 3, 5)**
ii. **(2, 3, 7)**

Hence even **both statements together are not sufficient**.

Subject:) CSAT
Topic:) Quantitative Aptitude
Subtopic:)

Q.48) A Question is given followed by two Statements I and II. Consider the Question and the Statements.
Question: Is (x + y) an integer?
Statement I: (5x + 3y) is an integer.
Statement II: (2x + y) is an integer.
Which one of the following is correct in respect of the above Question and the Statements?
a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone
b) The Question can be answered by using either Statement alone
c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone
d) The Question cannot be answered even by using both the Statements together

Ans) c
Exp) Option c is the correct answer.
Check Statement I alone
Given:
5x + 3y is an integer.
This alone does **not** guarantee that **x + y** is an integer.
Example 1: Let **x = 1/2, y = ½**
Then: **5x + 3y = 5/2 + 3/2 = 4**, which is an integer
and **x + y = 1**, which is an integer
Example 2: Let **x = 1/5, y = 0**
Then: **5x + 3y = 1**, which is an integer
but **x + y = 1/5**, which is **not** an integer
So **Statement I alone is not sufficient**.
Check Statement II alone
Given: **2x + y** is an integer.
This alone also does **not** guarantee that **x + y** is an integer.
Example 1: Let **x = 1, y = 0**
Then: **2x + y = 2**, which is an integer
and **x + y = 1**, which is an integer
Example 2: Let **x = 1/2, y = 0**
Then: **2x + y = 1**, which is an integer
but **x + y = 1/2**, which is **not** an integer
So **Statement II alone is not sufficient**.
Check both statements together
5x + 3y is an integer
2x + y is an integer
Now subtract the double of second expression from the first:
(5x + 3y) − 2 (2x + y) = x + y
So: **x + y** is an integer.
So **both statements together are sufficient**.
Subject:) CSAT
Topic:) Quantitative Aptitude
Subtopic:)

Directions for the following 2 (two) items:

Read the following passage and answer the items that follow the passage. Your answers to these items should be based on the passage only

Passage - 1

The rapid expansion of digital connectivity has transformed online gaming from a niche form of entertainment into a vast and influential digital ecosystem embedded in the daily lives of billions, especially adolescents and young adults. However, this growth has brought with it a range of serious concerns. Real money gaming platforms, betting applications, and unregulated gaming environments have created opportunities for organised criminal networks to exploit users through scams, deceptive practices, and even money laundering. At the same time, online gaming platforms are emerging as potential channels for extremist networking and radicalisation. Alongside these security risks, the psychological and social consequences of excessive gaming, such as addiction, social isolation, and declining mental well being, are increasingly disrupting real world relationships and overall quality of life. As an entire generation grows up immersed in these digital environments, the critical challenge for governments, technology companies, and civil society is to ensure that these platforms do not expose users to financial, psychological, and social harm.

Q.49) Which one of the following statements best reflects the most logical and rational inference that can be drawn from the passage?
a) By often doing more harm than good, online gaming platforms negatively impact users' well-being.
b) The expansion of internet connectivity has fueled the growth of online gaming platforms.
c) As online gaming continues to grow, we must ensure it does not harm users' well-being.
d) As children are more susceptible to the negative effects of online gaming, access should be restricted for them.

Ans) c
Exp) Option c is the correct answer.
Statement a is incorrect. While the passage states that excessive gaming could cause "addiction, social isolation, and declining mental well being", statement a does not best reflect the most logical and rational inference that can be drawn from the passage as it **narrowly focuses upon the harm caused by online gaming.**
Statement b is incorrect. While the idea is mentioned in the passage, statement b does not best reflect the most logical and rational inference that can be drawn from the passage as it **does not highlight the consequences of the growth of online gaming.**
Statement c is correct. It best reflects the most logical and rational inference that can be drawn from the passage as it comprehensively highlights the main message of the passage. It is mentioned in the passage as "**As an entire generation grows up immersed in these digital environments, the critical challenge for governments, technology companies, and civil society is to ensure that these platforms do not expose users to financial, psychological, and social harm**."
Statement d is incorrect. While **the passage states that adolescents and young adults are more attracted to online gaming, it does not explicitly state that access to online gaming should be restricted for them.** Therefore, statement d does not best reflect the most logical and rational inference that can be drawn from the passage.
Subject:) CSAT
Topic:) Reading Comprehension
Subtopic:)

Q.50) With reference to the above passage, the following assumptions have been made:
I. Online gaming platforms can be exploited to incite social unrest among communities.
II. Since games involving real money are susceptible to fraud, they should be banned.
Which of the above assumptions is/are valid?
a) I only
b) II only
c) Both I and II
d) Neither I nor II

Ans) a

Exp) Option a is the correct answer.

Statement I is correct. The passage states "**online gaming platforms are emerging as potential channels for extremist networking and radicalisation**". Therefore, we can conclude that assumption made in statement I is correct.

Statement II is incorrect. The passage states "Real money gaming platforms, betting applications, and unregulated gaming environments have created opportunities for organised criminal networks to exploit users through scams, deceptive practices, and even money laundering". However, **it cannot be assumed that online games involving money should be banned.**

Subject:) CSAT

Topic:) Reading Comprehension

Subtopic:)